Lierda NT26-FCN B Series

Hardware Design Manual

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Revision History

Document version	Change Date	Reviser	Reviewer	Change content
Rev1.0	23-10-30	CWY、CZT	CG、SLY、YHB、YMX	Initial version
Rev1.1	23-11-20	CWY、CZT	CG、SLY、YHB、YMX	Key feature added note: VDD_EXT non-volatile hardware version
Rev1.2	23-12-08	CWY、CZT	CG、SLY、YHB、YMX	Add descriptions for the SPK_P and SPK_N pins; Change ADC voltage range.
Rev1.3	24-03-12	CWY	YHB	Optimize the module dimension diagram and package diagram.
Rev1.4	24-04-26	CWY	YHB	Update the description of the MAIN_DTR pin.
Rev1.5	24-10-10	CZT	SLY	Add power domain description. GPIO power domain corresponds to VO_LDOIO; AGPIO power domain corresponds to LDO_AONIO; WAKEUP power domain corresponds to VDD18AON
Rev1.6	24-10-28	CZT	SLY	Increase the description, LDO_AONIO power domain voltage is software configurable; Add a note to avoid mercury or mercury vapor environment when using the module for production.
Rev1.7	25-03-03	CZT		Add applicable module model NT26FCNB70WNA
Rev1.8	25-03-19	CZT		Modify the shutdown precautions; Modify the ADC sampling range; Modify the working frequency band range of B41; Increase the module expansion power supply voltage; Add a section for special characters 1.1;

				The power on chapter is divided into the power on chapter and the power off chapter for separate descriptions.
Rev1.9	25-04-08	CWY		Modify the description of the reference layer in section 5.4 of the PCB. Add NT26FCNB60WNA module.
Rev2.0	25-04-29	CZT		Update power consumption description of sleep mode.

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Users are responsible for complying with relevant regulations on wireless communication modules and devices in other countries and specific environmental regulations. By following the following safety principles, personal safety can be ensured and help protect products and work environments from potential damage. Our company is not responsible for any losses resulting from customers' failure to comply with these regulations.

	Road safety comes first! When driving, please do not use handheld mobile devices unless they have a hands-free function. Please pull over before making a call!
	Please turn off your mobile devices before boarding. The wireless function of mobile devices is prohibited from being turned on in the airplane to prevent interference with the aircraft communication system. Ignoring this prompt may compromise flight safety and even violate the law.
	When in a hospital or healthcare facility, pay attention to whether there are restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so it may be necessary to turn off mobile terminal devices.
	Mobile terminal devices do not guarantee effective connection in all situations, such as when the mobile terminal device has no balance or the SIM card is invalid. In case of emergency situations when you encounter the above circumstances, please remember to use emergency calls, and ensure that your device is powered on and in an area with sufficient signal strength.
	Your mobile terminal device will receive and emit radio frequency signals when it is powered on, which may cause radio frequency interference when it is close to a TV, radio, computer, or other electronic devices.
	Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off your mobile terminal device. Operating electronic devices in any location with potential explosion hazards poses a safety risk.

Module selection for application

Serial number	Module model	Support frequency band	Dimensions (mm)	Module introduction
1	NT26FCNB10WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power failure
2	NT26FCNB30WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off Support VoLTE
3	NT26FCNB00WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off
4	NT26FCNB70WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power off Support VoLTE
5	NT26FCNB60WNA	Band1/3/5/8/ 34/38/39/40/41	17.7×15.8×2.4	VDD_EXT power-off

Content

1 Introduction

This document defines the application specifications of the Lierda NT26-FCN B series Cat.1 wireless communication module, describing its frequency band function, key features, hardware interface, operating modes, electrical characteristics, mechanical specifications, packaging storage, etc., to help users quickly grasp the application method of this Cat.1 module and facilitate fast, flexible product design.

For ease of description, the NT26-FCN B series Cat.1 wireless communication module mentioned in this document will be referred to as the "NT26-FCN B series module" uniformly.

1.1 Special Symbols

Table 1-1 Special Symbol Description

Symbol	Definition
※	If not specified, the asterisk (*) marked after the module functions, features, interfaces, pin names, AT commands, or parameters indicates that the function, feature, interface, pin name, AT command, or parameter of the module is under development and not supported temporarily.

2 Product Overview

The NT26-FCN B series module is a leading Cat.1 wireless communication module globally, compliant with 3GPP R14 Cat.1bis standard, supporting bandwidths of 1.4/3/5/10/15/20MHz, with features including small size, low power consumption, strong anti-interference capability, etc. The NT26-FCN B series module is suitable for various common IoT application scenarios, such as:

- Smart Meter Reading
- Intelligent Parking
- Smart City
- Smart Security

Asset Tracking

- Smart Home Appliances
- Agricultural and environmental monitoring

2.1 Frequency bands and functions

The wireless network functions supported by the module are as follows:

Table 2-1 Module Supports Wireless Network Functions

Network function	Explanation
LTE-FDD	B1/B3/B5/B8
LTE-TDD	B34/B38/B39/B40/B41
WIFI※	WIFI Scan

Attention

LTE-TDD B41 only supports 140M (2535-2675MHz).

2.2 Key features

The key features of the module are as follows:

Table 2-2 Module Key Features Description

Parameters	Explanation
Module packaging	LGA
Module size	17.7mm×15.8mm×2.4mm (L×W×H)
Module weight	About 1.5 grams
Operating voltage	◆ Supply Voltage: 3.3~4.5V, Typical Value: 3.8V ◆ Extended power supply voltage(1): 3.1～4.5V
VDD_EXT characteristics	The B series offers two hardware versions. Sleep mode: VDD_EXT power off (default) Sleep mode: VDD_EXT not power off (customized)
Ultra low power	Sleep mode: 3uA@3.8V (typical value)3uA@3.8V Partial model 8.5uA@3.8V, see section 6.3 for power consumption.8.5uA@3.8V
Transmit Power	23dBm±2.7dB（Max）
Antenna interface	50 ohm characteristic impedance
USIM	Supported USIM card types: Class B (3.0 V) and Class C (1.8 V)
UART	Main serial port: ◆ Used for AT command transmission and data transfer, the supported default baud rate is 115200bps. ◆ Used for firmware upgrade, the supported baud rate defaults to 921600bps
	Debug serial port: ◆ Used for software debugging. ◆ The default baud rate of the serial port is 3Mbps.
	Auxiliary serial port: ◆ User-defined function
USB	◆ Compatible with USB 2.0 (support host mode only), with a maximum data transfer rate of 480 Mbps. ◆ Used for AT command transmission, data transfer, software debugging, and firmware upgrade.

I2C	Support I2C function
SPI	Support SPI function
PCM	Support PCM function
SPK	Support SPK function
CAMERA	Software adaptation can support camera function.
LCD	Software adaptation can support LCD functionality.
Indicator light	◆ Running status indicator ◆ Network status indicator
Communication Interface Characteristics	Support 3GPP Rel.13/14 Cat.1 wireless communication interface and protocol.
Network Protocol Features	Support protocols such as TCP/UDP/HTTP(S)/SSL/MQTT(S)/FOTA/PPP/RNDIS/FTP/ECM.
Data transmission characteristics	◆ LTE-FDD: Maximum downlink speed of 10Mbps, maximum uplink speed of 5Mbps ◆ LTE-TDD: Maximum downlink speed of 8.96Mbps, maximum uplink speed of 3.1Mbps
Firmware upgrade	◆ Main serial port upgrade ◆ USB interface upgrade ◆ DFOTA upgrade
Operating temperature	◆ Normal operating temperature range(1): -35~75℃ ◆ Extended operating temperature range(2): -40~85℃
Storage temperature	Storage temperature range(3): -40~90℃
RoHS	All components comply with EU RoHS standards.

Note

(1) See section 6.2 for a specific description of the expansion voltage.

(2)(3)(4) See section 6.6 for specific description about temperature.

2.3 Function block diagram

NT26-FCN B series modules have rich peripheral interfaces and RF functions, the functional block diagram is as follows:

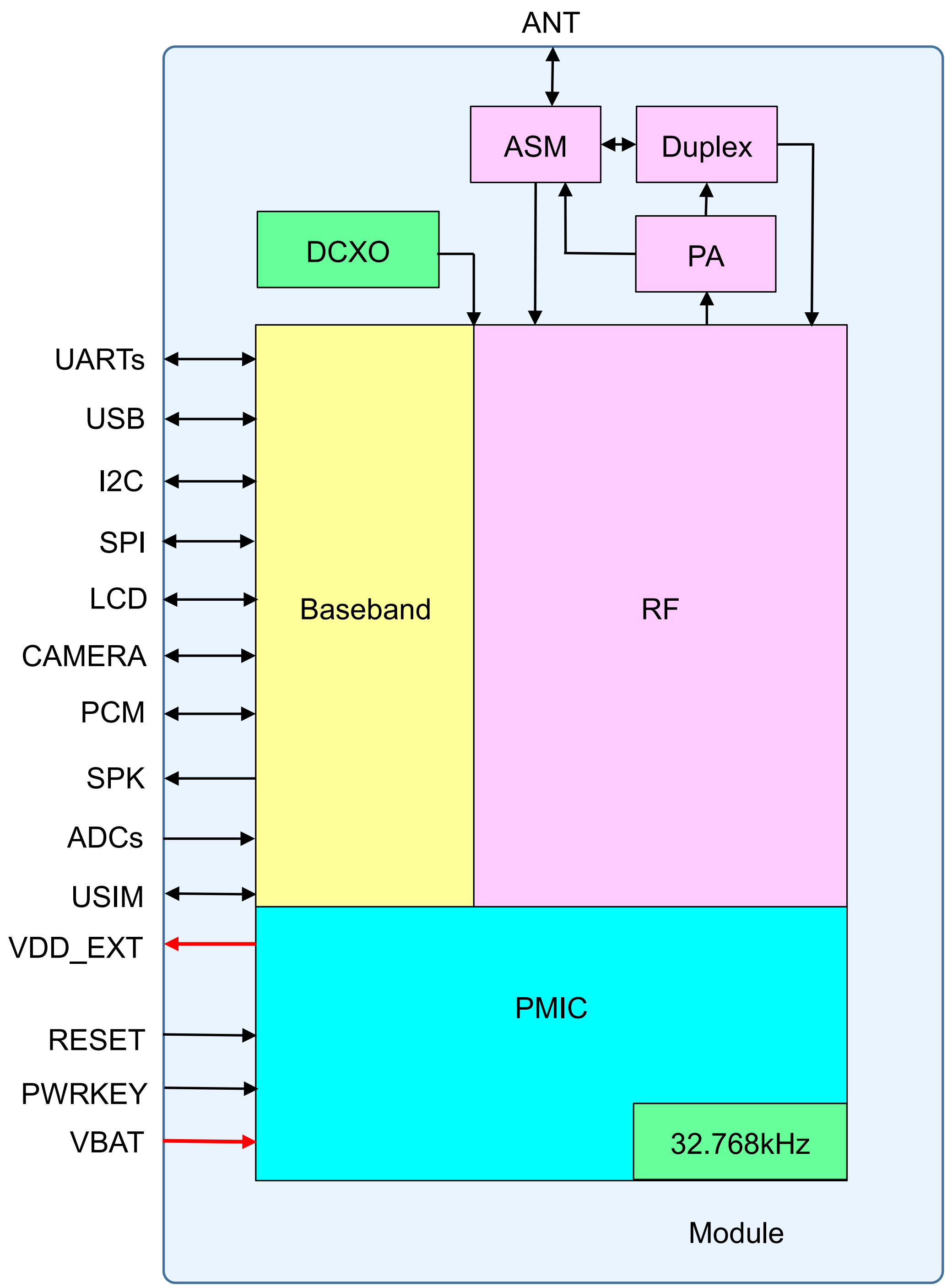


Figure 2-1 Module Functional Block Diagram

2.4 Pinout diagram

NT26-FCN B series module has a total of 109 pins, all of which are LGA pins. The pinout diagram of the module is as follows:

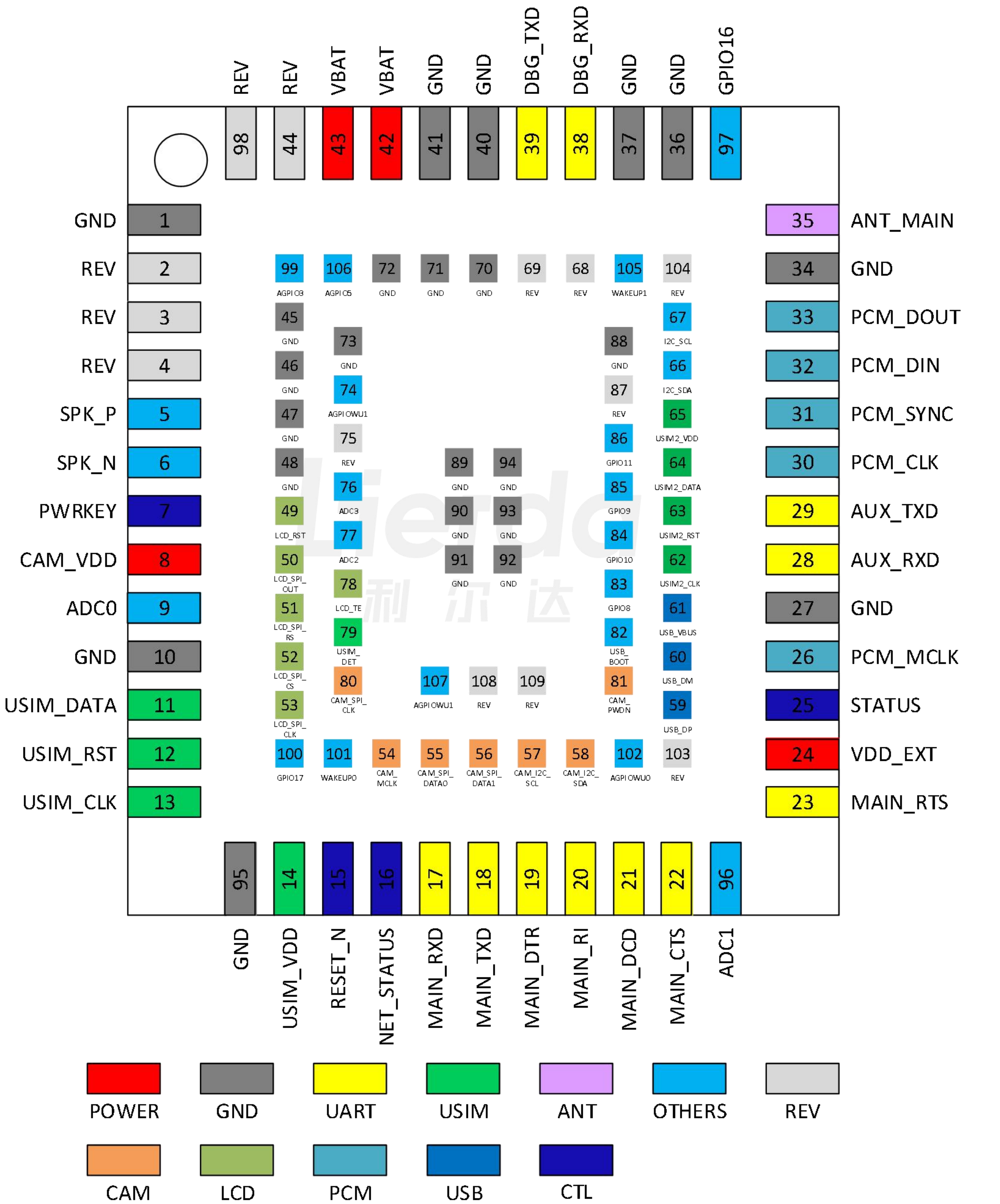


Figure 2-2 Module Pinout Diagram

2.5 Pin Description Table

To better understand the application, the following table provides a definition of the types of I/O parameters:

Table 2-3 Definition Explanation of I/O Parameter Types

I/O parameter types	Explanation
DI	Numeric input
DO	Number output
DIO	Numeric input and output
AI	Simulated input
AO	Simulated output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	GND

In order to better understand the application, the following table describes the characteristics of the power domain parameters:

Table 2-4 Module Power Domain Characteristics Description

Power domain parameters Type	DC characteristics	Explanation	Power supply object
VBAT	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	Module power input, recommended to use 3.8V/1.2A power supply.	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 1.8V; Different hardware versions: In sleep mode, VDD_EXT power-off version, driving capability is 120mA;	VDD_EXT

		In sleep mode, the non-power-off version of VDD_EXT has a driving capability of 3mA.	
VO_LDOIO	VOLmax=0.2×VO_LDOIO VOHmin=0.7×VO_LDOIO VILmax=0.15×VO_LDOIO VIHmin=0.8×VO_LDOIO	Default 1.8V, software configurable; Power off in sleep mode.	UART GPIO
VDD18AON	VILmax=0.2×VDD18AON VIHmin =0.7×VDD18AON VOLmax=0.15×VDD18AON VOHmin=0.8×VDD18AON	Not losing power in sleep mode.	WAKEUP RESET
LDO_AONIO	VILmax=0.2×LDO_AONIO VIHmin =0.7×LDO_AONIO VOLmax=0.15×LDO_AONIO VOHmin=0.8×LDO_AONIO	Default 1.8V, configurable by software; No power consumption in sleep mode.	AGPIO
VO_LDOSIM	Vnorm=1.8/3.0V VOLmax=0.15×VO_LDOSIM VOHmin=0.8×VO_LDOSIM VILmax=0.2×VO_LDOSIM VIHmin=0.7×VO_LDOSIM	SIM card dedicated power supply, supporting 1.8/3.0V cards	USIM

Table 2-5 Module Pin Description

1. Power					
Pin number	**Pin names**	**I/O**	**Description**	**DC characteristics**	**Note**
42,43	VBAT	PI	Power supply	Vmax=4.5V Vmin=3.3V Vnorm=3.8V	External power supply needs to provide a current carrying capacity of 1.2A or above.
24	VDD_EXT	PO	Power output	Vnorm=1.8V $I_O$max=50mA	Can be used for external circuit pull-up or reference level; default power-off in sleep mode; leave floating if not used, it is recommended

					to reserve test points.
1,10,27,34,36, 37,40,41,45~48, 70~73,88~95	GND	G	Ground	-	

2. Power on/off and reset					
Pin number	Pin names	I/O	Describe	DC characteristics	Note
7	PWRKEY	DI	Module power on/off pins	$V_{IL}max=0.45V$	Internal pull-up, low level effective, if not used, leave floating
15	RESET_N	DI	Module reset pin	$V_{IL}max=0.45V$	Internal pull-up, low level effective, if not used, leave floating.

3、USB					
Pin number	Pin name	I/O	Description	DC Characteristics	Note
59	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
60	USB_DM	DIO	USB differential data (-)	-	
61	USB_VBUS	DI	USB_VBUS input wake-up	-	

4. Serial Port UART					
Pin numbe	Pin names	I/O	Description	DC characterist	Note

r				ics	
17	MAIN_RXD	DI	Main serial port receives data	VO_LDOIO	
18	MAIN_TXD	DO	Main serial port sends data	VO_LDOIO	
19	MAIN_DTR	DI	Main serial port data terminal ready	LDO_AONIO/ VDD18AON	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep
20	MAIN_RI	DO	Main serial port outputs ringing prompt	LDO_AONIO	
21	MAIN_DCD	DO	Main serial port output carrier detection	VO_LDOIO	
22	MAIN_CTS	DO	Main serial port DTE clear to send	VO_LDOIO	Reuse SPK_P
23	MAIN_RTS	DI	Master serial port DTE requests transmission.	VO_LDOIO	Reuse SPK_N
38	DBG_RXD	DI	Debugging serial port receiving data	VO_LDOIO	
39	DBG_TXD	DO	Debug serial port to send data	VO_LDOIO	
28	AUX_RXD※	DI	Assist in receiving data through the serial port.	VO_LDOIO	Not supported at the moment.
29	AUX_TXD※	DO	Assist in sending data through the serial port.	VO_LDOIO	Not supported at the moment.

5. USIM card interface					
Pin number	Pin names	I/O	Description	DC characteristics	Note
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DIO	SIM card reset line	VO_LDOSIM	
13	USIM_CLK	DIO	SIM card clock line	VO_LDOSIM	
14	USIM_VDD	PO	SIM card power	VO_LDOSIM	1.8/3.0V
79	USIM_DET	DI	SIM card status signal detection	VDD18AON	High level effective
62	USIM2_CLK	DIO	SIM card 2 clock line	VO_LDOIO	USIM2 pin is multiplexed with CAMERA section pins
63	USIM2_RST	DIO	SIM card 2 reset line	VO_LDOIO	
64	USIM2_DATA	DIO	SIM card 2 data cable	VO_LDOIO	
65	USIM2_VDD	PO	SIM card 2 power	1.8/3.0V $I_O max=50mA$	Powered by the same power source as USIM_VDD.

6. Antenna RF interface					
Pin number	Pin names	I/O	Description	DC characteristics	Note
35	ANT_MAIN	AIO	RF antenna interface	50 ohm characteristic impedance	

7. Camera interface					
Pin	Pin names	I/O	Description	DC	Note

number				characteristics	
8	CAM_VDD	PO	Camera power supply	VO_LDOIO	If not used, leave floating
54	CAM_MCLK	DO	Camera main clock	VO_LDOIO	If not used, leave floating
55	CAM_SPI_DATA0	DI	Camera SPI data bit 0	VO_LDOIO	If not used, leave floating
56	CAM_SPI_DATA1	DI	Camera SPI data bit 1	VO_LDOIO	If not used, leave floating
57	CAM_I2C_SCL	DO	Camera I2C clock	VO_LDOIO	Recommend external pull-up, leave floating if not used.
58	CAM_I2C_SDA	DIO	Camera I2C data	VO_LDOIO	Recommend external pull-up, leave floating if not used.
80	CAM_SPI_CLK	DI	Camera SPI clock	VO_LDOIO	If not used, leave floating
81	CAM_PWDN	DO	Camera turned off.	VO_LDOIO	If not used, leave floating

8. LCD Interface					
Pin number	**Pin names**	**I/O**	**Description**	**DC characteristics**	**Note**
49	LCD_RST	DO	LCD reset	VO_LDOIO	If not used, leave floating

50	LCD_SPI_DOUT	DIO	LCD data output	VO_LDOIO	If not used, leave floating
51	LCD_SPI_RS	DO	LCD register selection	VO_LDOIO	If not used, leave floating
52	LCD_SPI_CS	DO	LCD chip select	VO_LDOIO	If not used, leave floating
53	LCD_SPI_CLK	DO	LCD clock	VO_LDOIO	If not used, leave floating
78	LCD_TE	DI	Lierda frame synchronization	LDO_AONIO	If not used, leave floating

9. I2C Interface					
Pin number	**Pin names**	**I/O**	**Description**	**DC characteristics**	**Note**
66	I2C_SDA	DIO	I2C serial data	VO_LDOIO	If not used, leave floating
67	I2C_SCL	DO	I2C serial clock	VO_LDOIO	If not used, leave floating

10. PCM Interface					
Pin number	**Pin names**	**I/O**	**Description**	**DC Characteristics**	**Note**
30	PCM_CLK	DO	PCM clock output	VO_LDOIO	If not used, leave floating
31	PCM_SYNC	DO	PCM synchronization signal	VO_LDOIO	If not used, leave floating

32	PCM_DIN	DI	PCM data input	VO_LDOIO	If not used, leave floating
33	PCM_DOUT	DO	PCM data output	VO_LDOIO	If not used, leave floating
26	PCM_MCLK	DO	PCM reference clock	VO_LDOIO	If not used, leave floating

11. SPK interface					
Pin number	Pin names	I/O	Description	DC characteristics	Note
5	SPK_P	DO	Audio differential output.	VO_LDOIO	Reuse MAIN_CTS
6	SPK_N	DO	Audio negative differential output	VO_LDOIO	Reuse MAIN_RTS

<table>
<tr><th colspan="6">12, ADC interface</th></tr>
<tr><th>Pin number</th><th>Pin name</th><th>I/O</th><th>Description</th><th>DC characteristics</th><th>Note</th></tr>
<tr><td>9</td><td>ADC0</td><td>AI</td><td>ADC0 interface</td><td rowspan="4">Internal direct connection
Input voltage range:
0~1.5V
Internal partial pressure
Input voltage range:
0~3.3V
Default 1.5V
Internal partial pressure can be activated through the software.</td><td>If not used, leave floating</td></tr>
<tr><td>76</td><td>ADC3</td><td>AI</td><td>ADC3 interface</td><td>If not used, leave floating</td></tr>
<tr><td>77</td><td>ADC2</td><td>AI</td><td>ADC2 interface</td><td>If not used, leave floating</td></tr>
<tr><td>96</td><td>ADC1</td><td>AI</td><td>ADC1 interface</td><td>If not used, leave floating</td></tr>
</table>

13. Status Indicator					
Pin number	Pin names	I/O	Description	DC properties	Note
16	NET_STATUS	DO	Network status indicator	LDO_AONIO	If not used, leave floating
25	STATUS	DO	Running status indicator	LDO_AONIO	If not used, leave floating

14. Other interfaces					
Pin number	Pin names	I/O	Description	DC characteristics	Note
82	USB_BOOT	DI	Emergency download mode control. Pull up to VDD_EXT before powering on the module, and the module will enter emergency download mode after startup.	VO_LDOIO	The default setting inside the module is weak pull-down, it is recommended to float.
97	GPIO16	DIO	GPIO, can be reused as SWCLK0	VO_LDOIO	If not used, leave floating
100	GPIO17	DIO	GPIO, can be reused as SWDIO0	VO_LDOIO	If not used, leave floating
102	AGPIOWU0	DIO	AGPIO function, power domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.	LDO_AONIO /VDD18AON	If not used, leave floating
74,107	AGPIOWU1	DIO	AGPIO function, power	LDO_AONIO	If not used,

			domain LDO_AONIO, no power loss during sleep. WAKEUP function, power domain VDD18AON, no power loss during sleep.	/VDD18AON	leave floating
99	AGPIO3	DIO	AGPIO, sleep mode does not consume power.	LDO_AONIO	If not used, leave floating
106	AGPIO5	DIO	AGPIO, sleep will not consume power.	LDO_AONIO	If not used, leave floating
101	WAKEUP0	AI	Awakening input	VDD18AON	If not used, leave floating
105	WAKEUP1	AI	Awaken input	VDD18AON	If not used, leave floating
83	GPIO8	DIO	GPIO, can be reused as SPI_CS	VO_LDOIO	If not used, leave floating
84	GPIO10	DIO	GPIO, can be reused as SPI_MISO	VO_LDOIO	If not used, leave floating
85	GPIO9	DIO	GPIO, can be reused as SPI_MOSI	VO_LDOIO	If not used, leave floating
86	GPIO11	DIO	GPIO, can be reused as SPI_SCLK	VO_LDOIO	If not used, leave floating

15. Reserve interface					
Pin number	**Pin names**	**I/O**	**Description**	**DC Characteristics**	**Note**
2~4,26,44,68,69,75,87,98, 108~109	REV	-	-	-	Undefined pin, recommend floating

2.6 Assessment suite

Lierda can provide a complete evaluation and development kit, including module adapter boards, Cat.1 & NB-IoT module universal development boards, etc. Please feel free to contact us for inquiries.

3 Working characteristics

3.1 Working mode

The NT26-FCN B series module has multiple operating modes. The following table briefly describes several common operating modes of the module:

Table 3-1 Working Modes

Pattern	Status Description
Normal working mode	Data transmission: The network connection is normal. The module power consumption depends on the network settings and data transmission rate.
	Idle status: The software is running normally. The module is connected to the network but no data interaction with the network.
Minimum functionality mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card are not working.
Sleep mode	The module's power consumption will be reduced to a minimum, but the module can still receive paging, short messages, and TCP/UDP data.
Shutdown mode	VBAT power supply does not disconnect, software stops working

Note

Please refer to the AT command data manual for related AT+CFUN commands.

3.2 Sleep mode

Modules can enter sleep mode via AT commands. In sleep mode, the module can reduce power consumption to a very low level while still being able to receive network information via RF.

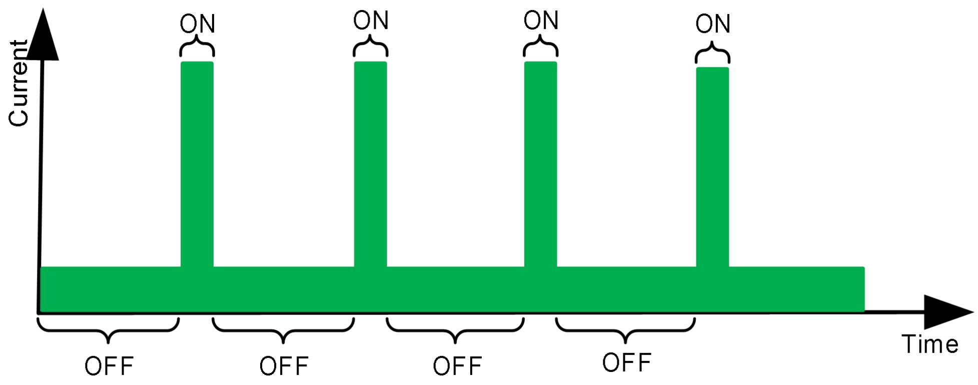


Figure 3-1 Power Consumption State of the Module in Sleep Mode

In sleep mode, the module peripherals are powered off to reduce power consumption, but the RF is in intermittent reception state and can still receive network information. When the module has a URC to report, it will wake up the host through the MAIN_RI pin.

3.2.1 Main serial port application scenarios

If the module and the host communicate via UART, two conditions must be met simultaneously to put the module into sleep mode:

- ◆ Enable sleep mode by executing the command AT+QSCLK=1;
- ◆ Ensure that MAIN_DTR remains at a high level or floating.

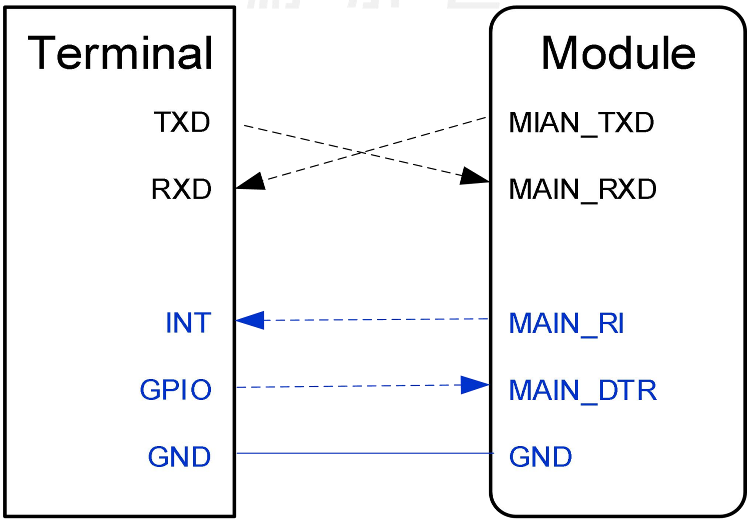


Figure 3-2 Application Circuit of Sleep Mode UART

- ◆ When the module needs to report URC, the module notifies the host through MAIN_RI.
- ◆ The host pulling down MAIN_DTR can wake up the module.

◆ If the host keeps pulling down MAIN_DTR, it can prevent the module from entering sleep mode.

If the sleep function is enabled using AT+QSCLK=2, the module can be woken up by sending any command to the main UART interface.

3.2.2 USB application scenarios

In the USB application scenario, the module needs to meet the following 3 conditions to enter sleep mode:

◆ Perform AT+QSCLK=1;

◆ Ensure MAIN_DTR remains at a high level or floating;

◆ The host USB bus connected to the module's USB interface enters suspended state.

3.2.2.1 Support USB remote wake-up function

The host supports USB suspend and resume, as well as USB remote wake-up functionality.

Please refer to the diagram for the connection between the module and the host:

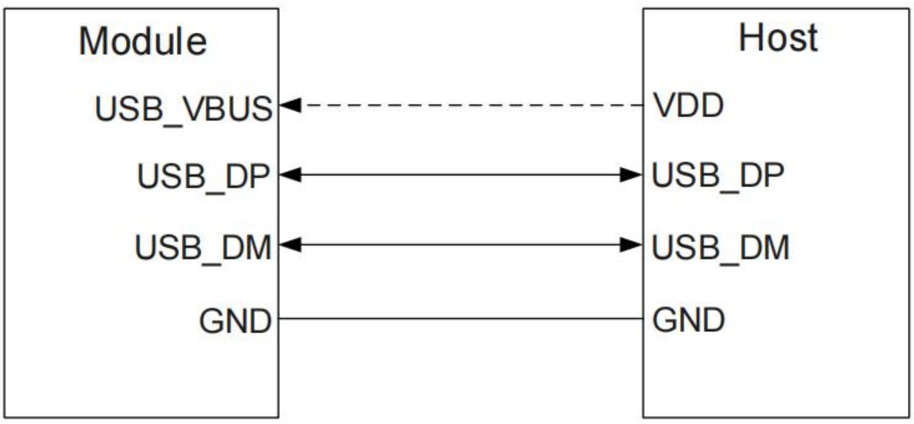


Figure 3-3 Sleep Application with USB Remote Wake-up Function

◆ Sending data to the module via USB will wake up the module.

When the module reports URC, the module will send a remote wake-up signal to wake up the host via the USB bus.

3.2.2.2 Support USB suspend, wake-up, and MAIN_RI functions.

If the host supports USB suspend and resume but does not support USB remote wake-up function, the host needs to be woken up by the MAIN_RI signal of the module.

The connection between the module and the host refers to the following diagram:

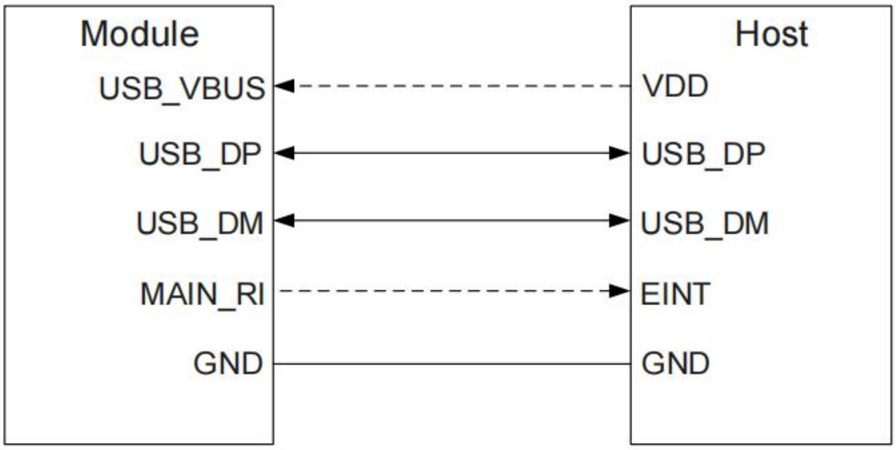


Figure 3-4 Sleep Application with MAIN_RI Function

- Sending data to the module via USB will wake up the module;
- When the module has URC reporting, the module can wake up the host through MAIN_RI.

3.2.2.3 Does not support USB suspend function.

When the host does not support USB suspend function, the module needs to meet the following 3 conditions to enter sleep mode:

- Execute AT+QSCLK=1;
- Ensure that MAIN_DTR remains at a high level or floating;
- Disconnect the USB_VBUS power supply.

If the host does not support USB suspend function, the module can enter sleep mode by disconnecting USB_VBUS through an external control circuit.

The connection between the module and the host is as shown in the diagram below:

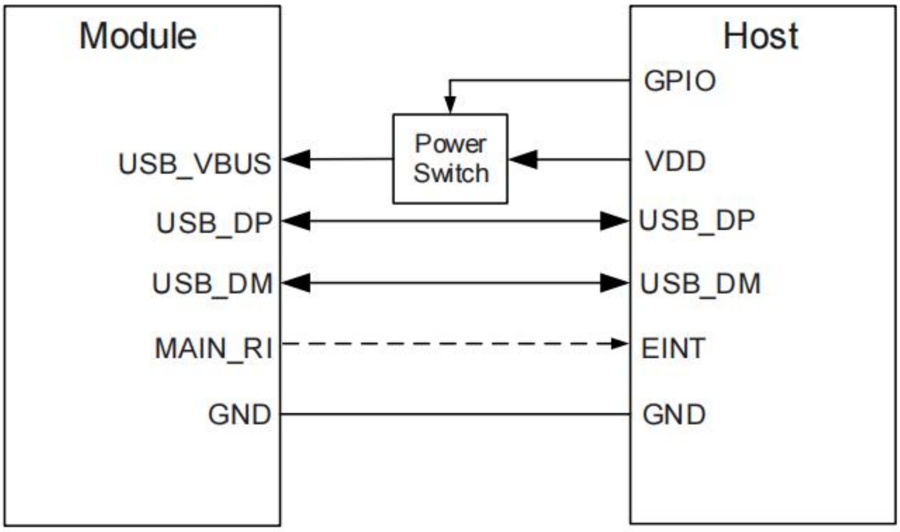


Figure 3-5 Sleep applications that do not support USB suspend functionality

Note

Restoring USB_VBUS power supply can wake up the module.

3.3 Flight mode

When the module enters flight mode, the RF function is not available, and all AT commands related to RF cannot be accessed. The module can be set to flight mode in the following ways:

The setting can be adjusted by sending the command AT+CFUN=<fun>. The <fun> parameter can be set to 0 or 1.

◆ AT+CFUN=0: Minimum functionality mode (disabling RF function and USIM card).

◆ AT+CFUN=1: Full functionality mode (default).

3.4 Power supply design

3.4.1 Introduction to the power supply

The NT26-FCN B series module provides 2 VBAT pins for connecting external power sources, and VDD_EXT is the power supply output of the module to the outside, which can be used as a reference voltage. The interface description is as follows:

Table 3-2 Power Supply Pin Definitions

Pin number	Pin name	Description	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Module power	3.3	3.8	4.5	V
24	VDD_EXT	Power on.	-	1.8	-	V

The VBAT power supply range of the module is 3.3 to 4.5V, and it is necessary to ensure that the lowest input voltage is not lower than 3.3V when the module is operating. The following figure illustrates the VBAT voltage drop situation of the module during sudden transmission:

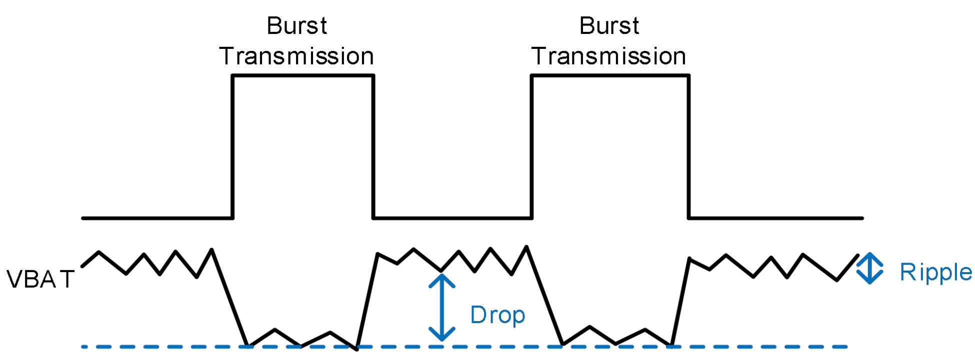


Figure 3-6 Emergency Transmission Power Requirements

Remarks

In sleep mode, VDD_EXT will be powered off.

3.4.2 Power supply design

When the module is operating at the maximum transmission power in the 4G network, the transient operating current under the current network condition can reach 1.2A, which may cause a drop in the power supply voltage. In any case, it is necessary to ensure that the module's power supply voltage remains above 3.3V, otherwise the module may experience unexpected conditions such as rebooting. To reduce voltage drops, it is recommended to use a 100uF filtering capacitor with low ESR (ESR <0.7Ω). It is also recommended to reserve three chip multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, it is recommended to add a high-power low-clamping voltage and high reverse pulse current Ipp TVS diode at the VBAT terminal to improve the module's surge voltage withstand capability.

The power reference circuit is as follows:

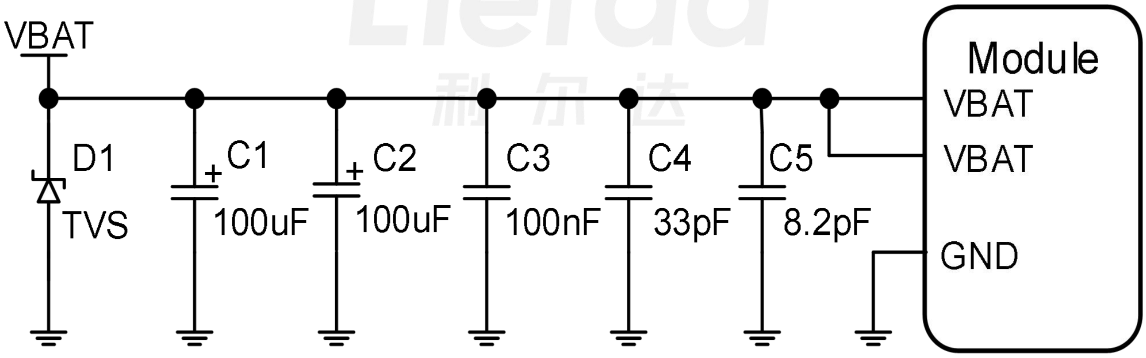


Figure 3-7 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace width. It is recommended that the trace width should not be less than 2mm. The GND plane of the power section should be as complete as possible and have multiple GND vias. Also, decoupling capacitors should be placed as close as possible to the VBAT pin of the module.

- ◆ D1 is a TVS diode with high power and low clamping voltage;
- ◆ C1 and C2 are 100uF low ESR tantalum capacitors, which improve the power supply's ripple current capability and stabilize the voltage.

◆ C3~C5 are ceramic packaged filtering capacitors of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.5 Power on.

3.5.1 Power-on introduction

The module can be powered on through the PWRKEY pin. The pin description is as follows:

Table 3-3 PWRKEY Pin Description

Pin number	Pin names	I/O	Description	Note
7	PWRKEY	DI	Module power on and off, low level effective	Default to high level.

The control logic of PWRKEY is as follows:

◆ The module can be powered on by pulling down PWRKEY for at least 500ms in the power-off state;

3.5.2 Reference design for power-on circuit

It is recommended to use an open-drain drive circuit or directly control the power on and off through a button. To prevent electrostatic discharge generated by contact, a TVS should be placed near the button for ESD protection. Please refer to the circuit diagram below:

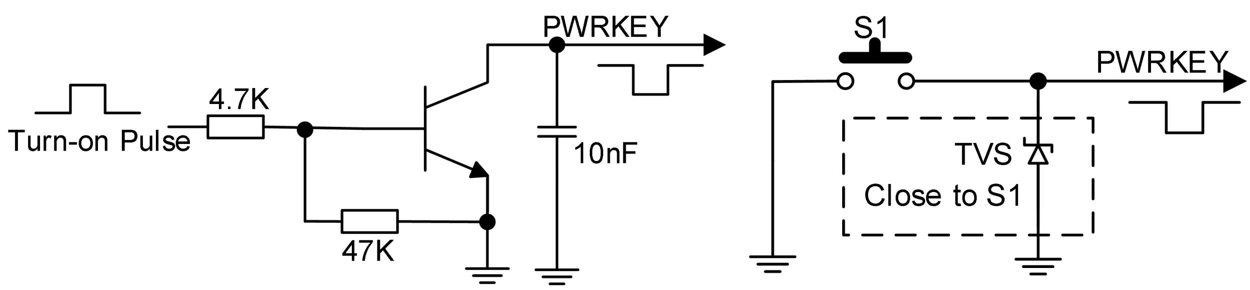


Figure 3-8 Power On/Off Circuit

3.5.3 Power-on sequence

The power-on sequence of the module is as shown in the following figure:

Figure 3-9 Power-On Sequence Diagram

Precautions

◆ The initial state of VBAT when powered on needs to be less than 0.1V, and the time for VABT to rise from 0V to 2.5V should be less than 10ms; before pulling down the PWRKEY pin, ensure that the VBAT voltage is stable, and it is recommended that the time interval between powering on VBAT and pulling down the PWRKEY pin is not less than 30ms.

◆ Pulling the USB_BOOT pin to VDD_EXT before powering on the module will force the module into emergency download mode upon startup.

◆ If automatic power-on is required without the need for shutdown function, PWRKEY can be directly pulled down to GND, and the pull-down resistor is recommended to be 4.7kΩ, or control PWRKEY to low level and keep it low with GPIO before powering on the module.

◆ The main serial port of the module sends the character "Lierda" to represent the completion of the boot.

3.6 Shutdown

The module can be shut down normally in the following ways:

◆ Power off the module by controlling it through the PWRKEY pin.

◆ Send the command AT+QPOWD to power off.

3.6.1 PWRKEY power off

In the power-on state, pull down PWRKEY for 650ms and then release it, the module will execute the shutdown process, the timing is as follows:

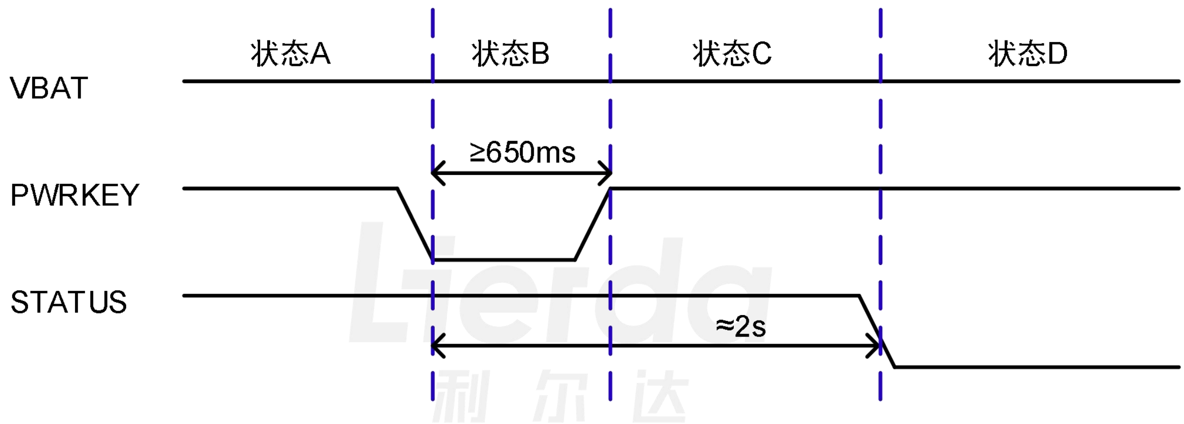


Figure 3-10 PWRKEY Shutdown Timing Diagram

PWRKEY shutdown timing description:

◆ Status A: Module working status;

◆ Status B: Lower the status of PERKEY, time should be at least 650ms or more;

◆ Status C: The module shuts down internally, from pulling down PWRKEY to complete shutdown takes about 2 seconds;

◆ Status D: The module has entered the shutdown state.

3.6.2 Shutdown using AT command

Executing AT+QPOWD can power off the module, this operation has the same timing and effect as powering off by pulling down PWRKEY.

Notes

◆ When the module is functioning normally, please do not cut off the power supply immediately, as it may damage the Flash data inside the module. It is recommended to first power off the module using the PWRKEY or AT command, and then disconnect the power supply.

◆ When using the AT command to shut down, please ensure that after the shutdown command is executed, the PWRKEY remains in a high level state. Otherwise, there may be a situation where the device powers on again after shutdown, which poses a risk of damaging the Flash if the power is disconnected at that time.

If the power-on is usually done by grounding the PWRKEY, it is recommended to use AT+QPOWD to execute the shutdown process before disconnecting the power.

3.7 Reset

3.7.1 Reset introduction

The RESET_N pin can be used for module reset. Pulling down the RESET_N pin for at least 300ms and then releasing it will reset the module. The description of the RESET_N pin is as follows:

Table 3-4 Reset Pin Description

Pin number	Pin names	I/O	Description	DC characteristics	Note
15	RESET_N	DI	Module reset/power off	VILmax=0.45V	There is a weak pull-up inside, low level effective, not connected can be left floating.

3.7.2 Reset circuit reference design

The RESET_N signal is sensitive to interference, it is recommended that the wiring of the module interface should be kept as short as possible and grounding should be handled. The typical reference applications of the hardware circuit are as follows, and the two circuit diagrams can be selected according to actual needs:

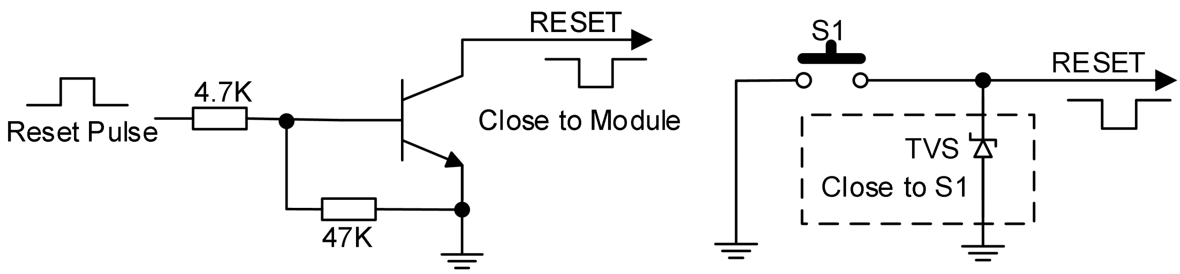


Figure 3-11 Reset Circuit Reference Design

3.7.3 Reset timing circuit

The reset timing sequence of the module is as follows:

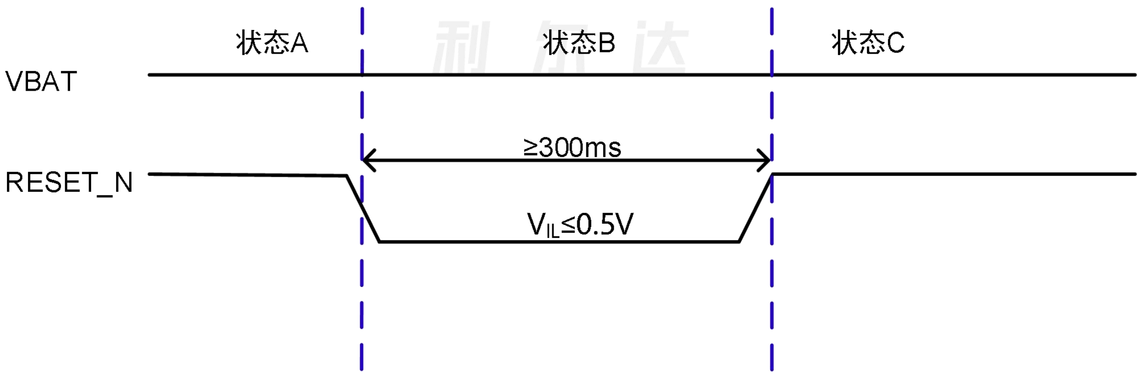


Figure 3-12 Reset Timing Diagram

Reset timing description:

- ◆ Status A: Module working normally;
- ◆ Status B: Pull down the RESET_N state for at least 300ms.
- ◆ Status C: The module is starting to reboot.

4 Application Interface

The NT26-FCN B series module provides a variety of peripheral functions, mainly including the following application interfaces:

- ◆ USB 2.0 interface
- ◆ UART interface
- ◆ USIM interface
- ◆ I2C interface
- ◆ PCM interface
- ◆ CAMERA interface
- ◆ LCD interface, etc.

4.1 USB interface

NT26-FCN B series modules support USB 2.0 interface, compliant with USB 2.0 specifications, and support Full-Speed (12 Mbps) and High-Speed (480 Mbps) modes. The module only supports USB slave mode. This interface can be used for AT command sending, data transmission, software debugging, and firmware upgrade. The interface definition is as follows:

Table 4-1 USB Interface Pin Description

Pin number	Pin names	I/O	Description	DC characteristics	Note
59	USB_DP	DIO	USB differential data (+)	-	If not used, leave floating
60	USB_DM	DIO	USB differential data (-)	-	If not used, leave floating
61	USB_VBUS	DI	Sleep wakefulness	VDD18AON	When in sleep mode, inserting a USB will

					wake up the module;
82	USB_BOOT	DI	USB mode configuration interface	VO_LDOIO	Internally weak pull-down, if not used, leave floating

4.1.1 USB circuit reference design

The module USB interface can be connected to either a USB connector or the MCU's USB interface. The reference schematic for connecting to the USB connector is as follows:

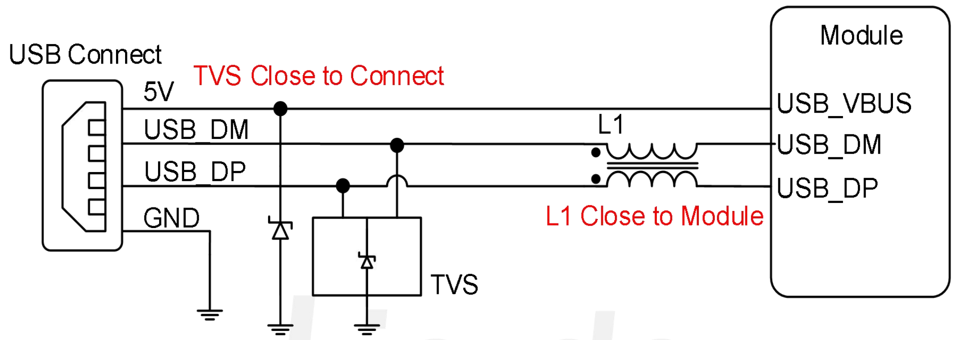


Figure 4-1 USB Interface Reference Design

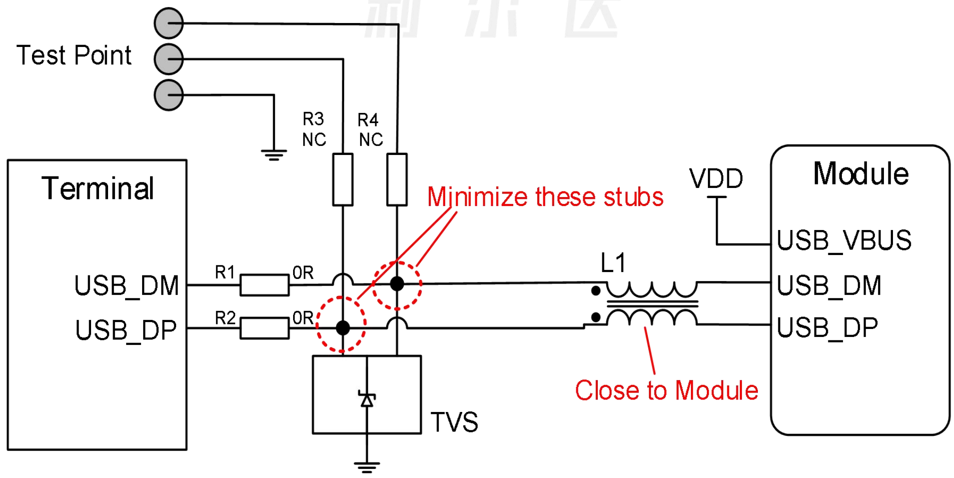


Figure 4-2 USB connection MCU reference design

Precautions

◆ To ensure the performance of USB, the routing of USB_DP and USB_DM should maintain a differential impedance of 90Ω, and ground treatment is needed around. It is recommended to connect a common mode inductor L1 between the MCU and the module to prevent USB signal from generating EMI interference, and L1 should be placed close to the module.

◆ USB traces should be kept away from crystal oscillators, oscillators, magnetic devices, RF signals, and strong signal areas. It is recommended to use inner layer differential traces and surround them with GND on all sides.

◆ If the module's USB interface is connected to the MCU through a connector, TVS diode protection should be added at the interface. Special attention should be paid to the selection of ESD protection devices, ensuring that the parasitic capacitance does not exceed 2pF, and they should be placed as close to the USB interface as possible.

- Reserve resistors R1~R4. Normally, connect R1~R2 without connecting R3~R4 to MCU. When debugging is needed, connect R3~R4 without connecting R1~R2 to switch to the test point. To meet the USB data line signal integrity requirements, keep the branch lines connecting to the test points as short as possible.

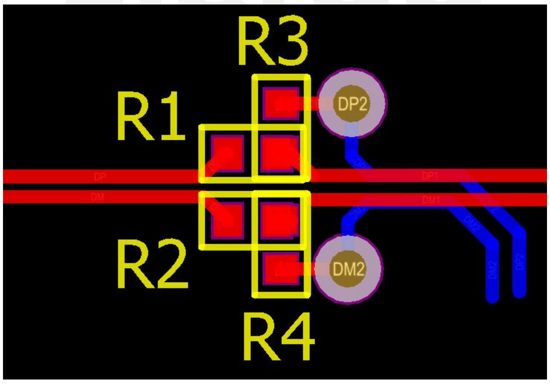


Figure 4-3 USB jumper routing diagram

4.1.2 USB data transfer

When the module is powered on or reset, if USB_BOOT is detected as a low level, the USB port will enumerate 2 serial ports for sending AT commands and logging output.

4.1.3 USB firmware download

When the module is powered on or reset, if USB_BOOT is detected as a high level, the USB port will enumerate a dedicated download port, and the download tool can upgrade the firmware through the USB interface. If no download tool is connected for more than 15 seconds, the module will re-enumerate 2 serial ports according to the normal USB configuration during startup.

The reference design for the USB_BOOT interface is as follows:

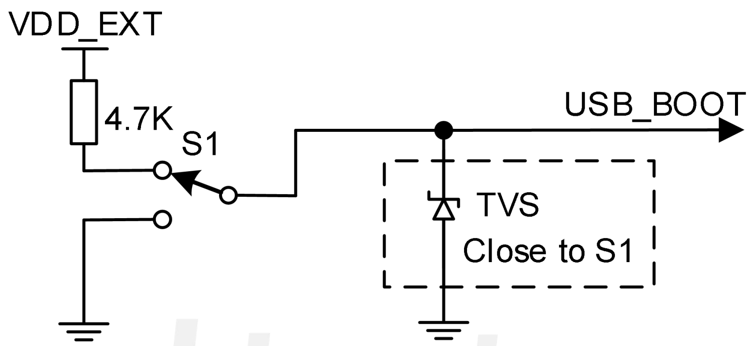


Figure 4-4 USB_BOOT Reference Design

4.2 UART communication

The module provides 3 general asynchronous transceivers: main serial port, debug serial port, and auxiliary serial port. The baud rate of the main serial port is configurable, and the debug serial port is only used for debugging and testing. The serial port pin definitions are as follows:

Table 4-2 Serial Port Pin Definitions

Interface	Pin number	Pin Names	Description	Note
Main Serial Port	17	MAIN_RXD	Main serial port receives data	Non-sleep mode: VDD_EXT power domain; Sleep mode: Switch to internal

				AON power domain.
	18	MAIN_TXD	Main serial port sends data	If not used, leave floating
	19	MAIN_DTR	Main serial port data terminal ready	In sleep mode, the wake-up pin can be used.
	20	MAIN_RI	Main serial port outputs ringing prompt.	If not used, leave floating
	21	MAIN_DCD	Main serial port output carrier detection	If not used, leave floating
	22	MAIN_CTS	Main serial port DTE clear to send	Connect CTS to DTE, leave floating if not used.
	23	MAIN_RTS	The main serial port DTE requests to send.	Connect RTS to DTE, leave floating if not used.
Debugging serial port	38	DBG_RXD	Debug serial port data reception	If not used, leave floating
	39	DBG_TXD	Debug serial port to send data	If not used, leave floating
Auxiliary serial port	28	AUX_RXD※	Assist in receiving data through the serial port.	Not supported at the moment.
	29	AUX_TXD※	Assist in sending data through the serial port.	Not supported at the moment

Precautions

◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 1.8V;

◆ In sleep mode, the VO_LDOIO power domain will power off, and MAIN_RXD will switch to the internal 1.2V power supply;

- The RXD pin of the serial port has an internal pull-up, so there is no need to connect an external pull-up resistor, and it has reverse connection protection function.

◆ Pay attention to the consistency of voltage levels when using the serial port, otherwise it may cause leakage current.

4.2.1 Serial port application

Main serial port characteristics:

◆ Used for AT command communication and data transmission, with a default baud rate of 115200bps;

◆ Configurable as: 4800/9600/19200/38400/57600/115200/230400/460800bps;

◆ Used for firmware upgrade, the default baud rate during upgrade is 921600bps;

◆ Support low power consumption wake-up module function.

Characteristics of serial port debugging:

◆ With the dedicated tools provided by the platform, you can view log information for software debugging, with a default speed of 3000000bps.

The schematic diagram of the serial port connection is as follows:

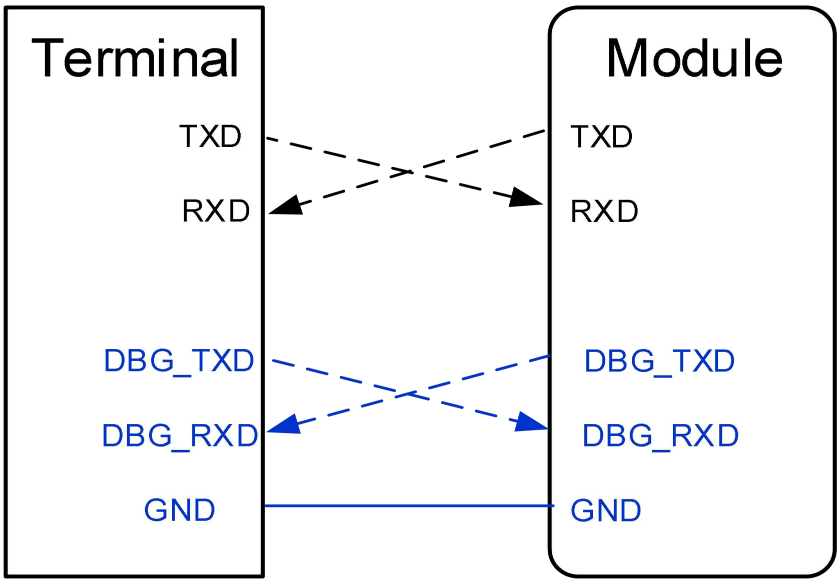


Figure 4-5 Serial Port Connection Diagram

4.2.2 Serial port circuit reference design

The main considerations for a suitable serial port level conversion circuit include whether it meets the working speed of the serial port, scenarios with low power requirements, and whether the power consumption meets the requirements, etc. It is recommended to refer to the following level matching circuit:

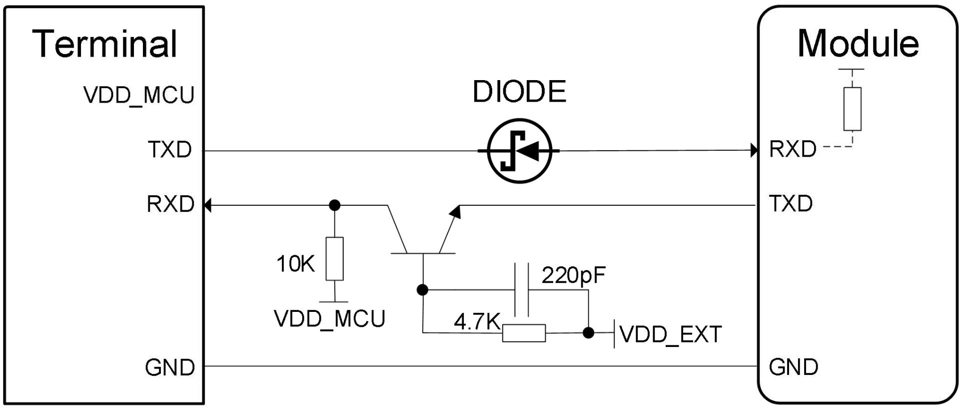


Figure 4-6 Transistor Level Conversion Reference Circuit

Recommend three transistors for reference:

Brand: CJ Model: S8050 J3Y Package: SOT-23

Recommend diodes, pay attention to the forward voltage of the diode ≤0.3V, for reference:

Brand: LRC Model: LRB520S-30T1G Package: SOD-523

4.3 USIM card interface

The module includes an external USIM card interface, compliant with ETSI and IMT-2000 standards, supporting 1.8V and 3.0V USIM cards. The pin descriptions are as follows:

Table 4-3 Definition of External USIM Card Interface Pins

Pin number	Pin names	Description	DC characteristics

11	USIM_DATA	SIM card data cable	VO_LDOSIM
12	USIM_RST	SIM card reset line	VO_LDOSIM
13	USIM_CLK	SIM card clock line	VO_LDOSIM
15	USIM_VDD	SIM card power supply	Vnorm=1.8/3.0V
91	USIM_DET	SIM card status signal detection	SIM card slot hot plug detection interface
62	USIM2_CLK	SIM card 2 clock line	VO_LDOIO
63	USIM2_RST	SIM card 2 reset line	VO_LDOIO
64	USIM2_DATA	SIM card 2 data cable	VO_LDOIO
65	USIM2_VDD	SIM card 2 power	Powered by the same source as USIM_VDD.

The module supports hot-plug detection of USIM card through USIM_DET, this function is enabled by default, supporting high/low level detection. If the USIM card detection function is not needed, USIM_DET can be left floating.

4.3.1 USIM card circuit reference design

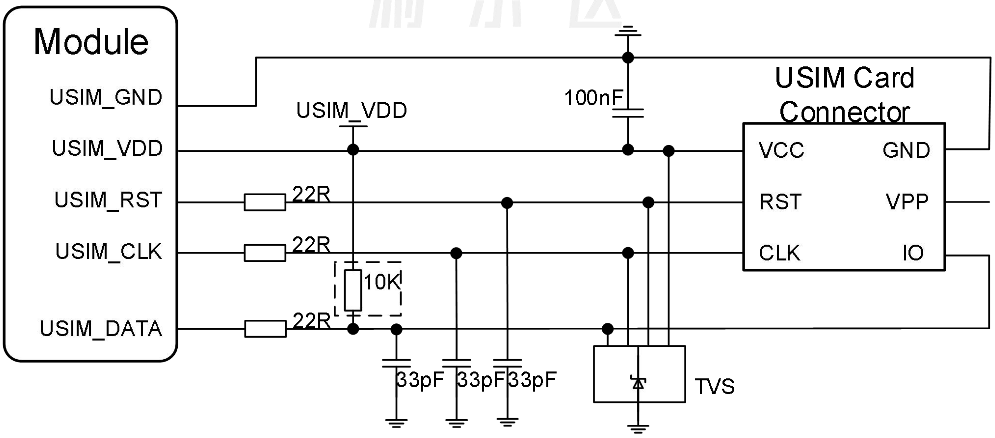


Figure 4-7 6PIN SIM card reference circuit

Precautions

The hardware design circuit of USIM2 can also refer to the above design, specifically refer to "Lierda NT26-FCN B Series Hardware Reference Design Manual".

4.3.2 Considerations for USIM card circuit design

To ensure the reliability and availability of the SIM card in the application, please follow the following USIM circuit design guidelines:

◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal line wiring length within 200mm if possible.

◆ SIM card signal line away from RF trace and VBAT power line;

◆ The GND wiring of the SIM card should be short and thick, ensuring that the wiring width is not less than 0.5mm;

◆ The decoupling capacitance of USIM_VDD should not exceed 1uF, and the capacitance should be placed close to the external SIM card's VCC.

◆ To avoid signal interference between USIM_DATA and USIM_CLK, their wiring should not be too close. Ground shielding should be added between the two traces. In order to prevent the impact of long traces, a pull-up resistor is generally added to USIM_DATA to USIM_VDD to improve the driving capability. If the traces are too long, it is recommended to externally reserve a 10K resistor near the card slot. In addition, the USIM_RST signal also needs to be protected by grounding.

◆ To ensure good ESD protection performance, it is recommended to add TVS diodes to the pins of the external USIM card. The parasitic capacitance of the selected TVS diodes should not exceed 15pF; a 22 ohm resistor should be placed in series between the signal lines of the module and the SIM card to suppress stray EMI and enhance ESD protection. Additionally, a parallel 33pF capacitor should be used to filter out RF interference. The relevant resistors, capacitors, and TVS diodes should be placed near the USIM card slot.

4.4 I2C interface

The module provides one I2C interface, and the interface description is as follows:

Table 4-4 I2C Interface Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
66	I2C_SDA	I2C data	VO_LDOIO	External pull-up required
67	I2C_SCL	I2C clock	VO_LDOIO	External pull-up required

4.5 PCM interface

4.5.1 Introduction to PCM

The module provides 1 PCM interface, the interface description is as follows:

Table 4-5 PCM Interface Pin Definitions

Pin number	Pin name	Description	DC Characteristics	Note
30	PCM_CLK	PCM clock output	VO_LDOIO	
31	PCM_SYNC	PCM synchronization signal	VO_LDOIO	
32	PCM_DIN	PCM data input	VO_LDOIO	
33	PCM_DOUT	PCM data output	VO_LDOIO	
26	PCM_MCLK	PCM reference master clock	VO_LDOIO	

4.5.2 Codec reference application

By using the PCM and I2C of the module, the Codec function can be implemented. The common application reference circuit is as follows:

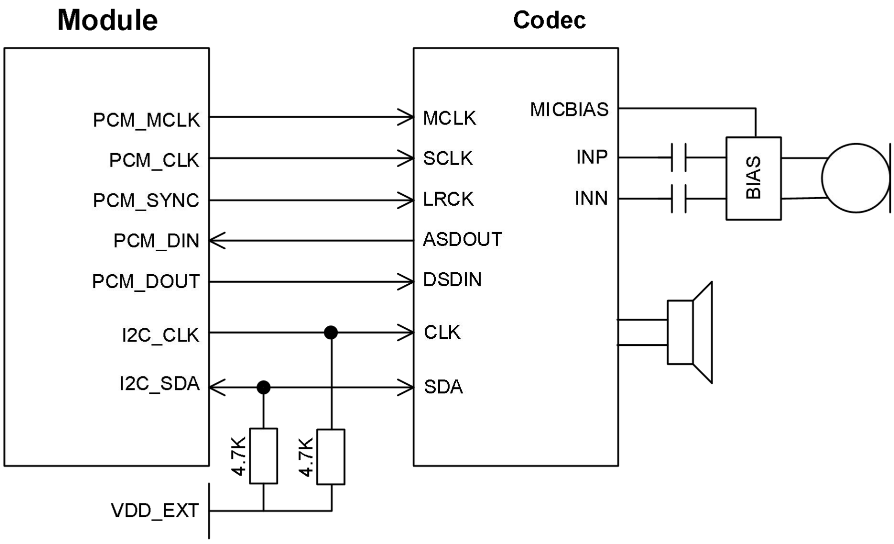


Figure 4-8 Codec Application Reference Circuit

Notes

"The Lierda NT26-FCN A Series Hardware Reference Design Manual provides various Codec design options, allowing users to choose the most suitable one according to their actual application."

4.6 SPK interface

4.6.1 Introduction to SPK

Table 4-6 PCM Interface Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
5	SPK_P	Audio differential output	VO_LDOIO	
6	SPK_N	Audio negative differential output	VO_LDOIO	

4.6.2 SPK Reference Application

By connecting the audio PA to the SPK interface of the module, the audio playback function can be achieved. The common application reference circuit is as follows:

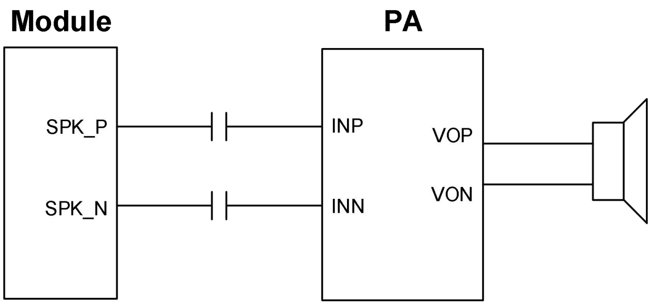


Figure 4-9 SPK Application Reference Circuit

The SPK function of the module, the actual circuit will vary depending on the selected PA, refer to the specific circuit in the "Lierda NT26-FCN B Series Hardware Reference Design Manual" document.

4.7 SPI and LCD interface

4.7.1 Interface Introduction

The module provides a set of LCD interfaces based on SPI. PIN[50..53] is a standard SPI interface group, which can also be reused for LCD functions in conjunction with other pins. The interface description is as follows:

Table 4-7 SPI Interface Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
49	LCD_RST	LCD reset	VO_LDOIO	
50	LCD_SPI_DOUT	Lierda data output	VO_LDOIO	Can be reused as SPI_MOSI
51	LCD_SPI_RS	LCD register selection	VO_LDOIO	Can be reused as SPI_MISO

52	LCD_SPI_CS	Lierda chip selection	VO_LDOIO	Can be reused as SPI_CS
53	LCD_SPI_CLK	LCD clock	VO_LDOIO	Can be reused as SPI_CLK
78	LCD_TE	Lierda frame synchronization	VO_LDOIO	

4.7.2 SPI Reference Application

The SPI of the module can be used as a master device or as a slave device. The application reference circuit is as follows:

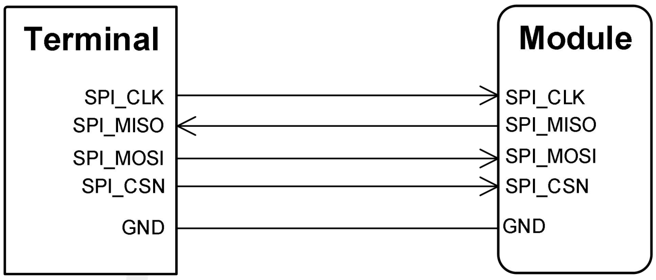


Figure 4-10 SPI Interface Circuit Reference Design (Module as Slave Device)

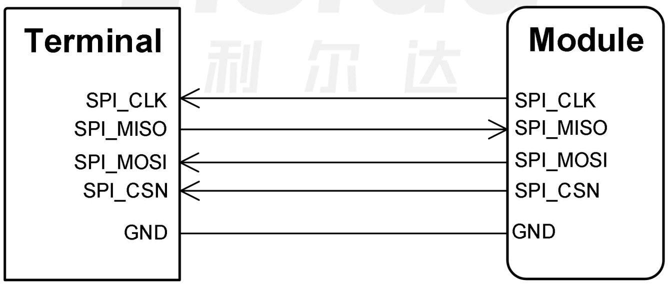


Figure 4-11 SPI interface circuit reference design (module as master device)

4.7.3 LCD reference application

The LCD function of the module, the actual circuit will vary depending on the selected LCD, please refer to the "Lierda NT26-FCN B Series Hardware Reference Design Manual" for specific circuit details.

4.8 Camera interface

4.8.1 Interface Introduction

The module provides a set of camera interfaces based on SPI and I2C, supporting up to 30-megapixel sensors, and allowing SPI single data line or dual data line transmission. The interface specifications are as follows:

Table 4-8 Camera Interface Pin Definitions

Pin number	Pin name	Description	DC Characteristics	Note
8	CAM_VDD	Camera power supply	VO_LDOIO	The module will lose power when it is in sleep or shutdown mode, and will be left floating if not used.
54	CAM_MCLK	Camera Master Clock	VO_LDOIO	If not used, leave floating
55	CAM_SPI_DATA0	Camera SPI data bit 0	VO_LDOIO	If not used, leave floating
56	CAM_SPI_DATA1	Camera SPI data bit 1	VO_LDOIO	If not used, leave floating
57	CAM_I2C_SCL	Camera I2C clock	VO_LDOIO	If not used, leave floating
58	CAM_I2C_SDA	Camera I2C data	VO_LDOIO	If not used, leave floating
80	CAM_SPI_CLK	Camera SPI clock	VO_LDOIO	If not used, leave floating
81	CAM_PWDN	Camera off	VO_LDOIO	If not used, leave floating

Notes

If the customer needs to power the camera with CAM_VDD, the following two solutions can be implemented:

- Modify VO_LDOIO to 2.8V through the software.
- The module hardware reserves a 2.8V LDO (not currently soldered), please contact Lierda to obtain this version of the module.

4.8.2 Camera reference application

The camera function of the module, the actual circuit will vary depending on the selected camera, please refer to the specific circuit in the "Lierda NT26-FCN B Series Hardware Reference Design Manual" document.

4.9 ADC interface

The module provides 4 12-bit analog-to-digital converter input interfaces (ADC interfaces). When wiring the ADC interfaces, to improve the voltage measurement accuracy of the interfaces, it is recommended to perform grounding. The pin definitions are as follows:

Table 4-9 ADC Pin Definitions

Pin number	Pin name	Description	DC characteristics	Note
9	ADC0	ADC0 interface	Internal direct connection Input voltage range: 0~1.5V Internal partial pressure Input voltage range: 0~3.3V Default 1.5V Internal partial pressure can be activated through software.	If not used, leave floating
76	ADC3	ADC3 interface		
77	ADC2	ADC2 interface		
96	ADC1	ADC1 interface		

Notes

If the collected voltage is greater than 3.3V, it is recommended to use a resistor divider circuit input for the ADC pin. The resistance value of the voltage divider resistor is recommended to be between 100 kΩ and 1MΩ, and it is recommended to use resistors with an accuracy of 1% or higher, otherwise it will reduce the measurement accuracy of the ADC.

◆ Reserve a 1nF capacitor at both ends of the ground potential divider resistor during the design, by default, it is not soldered.

4.10 Status indicator signal

The module has 1 network status indicator and 1 running status indicator interface, the interface definitions are as follows:

Table 4-10 Definition of Status Indicator Pins

Pin number	Pin name	Describe	DC characteristics	Note
16	NET_STATUS	Network status indicator	VDD18AON	If not used, leave floating
25	STATUS	Running status indicator	VDD18AON	If not used, leave floating

4.10.1 Network status indicator

The NET_STATUS pin is used to indicate the network status of the module, the logic levels under different states are as follows:

Table 4-11 Definition of Status Indicator Pins

Pin name	Voltage level logic	Network status

NET_STATUS	Slow flash (200ms high/1800ms low)	Search network status
	Slow flash (1800ms high/200ms low)	Standby mode
	Quick Flash (125ms high/125ms low)	Data transmission mode

4.10.2 Running status indicator

The STATUS pin is used to indicate the operating status of the module. When the module is powered on normally, STATUS will output a high level, and it remains valid even in sleep mode.

4.10.3 Indicator Light Circuit Reference Design

NET_STATUS and STATUS pins can both be used to drive LED indicator lights, the reference design is as follows:

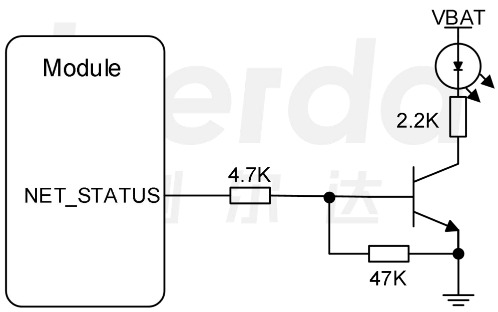


Figure 4-12 Indicator Light Reference Circuit

5 Radio Frequency Characteristics

5.1 Introduction to RF Function

NT26-FCN B series modules provide a main antenna RF interface, through which all air interface data is transmitted or received. The interface impedance is 50Ω. Interface description is as follows:

Table 5-1 RF Interface Pin Definitions

Pin number	Pin name	I/O	Description	DC characteristics	Note
35	ANT_MAIN	AIO	Main antenna interface	-	Used for air interface data transmission

5.2 Cellular network

5.2.1 Working frequency band

The operating frequency bands supported by the NT26-FCN B series module are as follows:

Table 5-2 Working Frequency Bands

Frequency band Band	Uplink frequency band Uplink(UL)band	Downlink frequency band Downlink(DL)band	Network standard Duplex Mode
1	1920MHz-1980MHz	2110MHz-2170MHz	FDD
3	1710MHz-1785MHz	1805MHz-1880MHz	FDD
5	824MHz-849MHz	869MHz-894MHz	FDD
8	880MHz-915MHz	925MHz-960MHz	FDD
34	2010MHz-2025MHz	2010MHz-2025MHz	TDD

38	2570MHz-2620MHz	2570MHz-2620MHz	TDD
39	1880MHz-1920MHz	1880MHz-1920MHz	TDD
40	2300MHz-2400MHz	2300MHz-2400MHz	TDD
41	2535MHz-2675MHz	2535MHz-2675MHz	TDD

Notes

LTE-TDD B41 only supports 140M (2535-2675MHz).

5.2.2 Transmit Power

The transmission power of the NT26-FCN B series module is as follows:

Table 5-3 Conduction Power

Band	Network standard	Maximum value	Minimum value	Note
1	FDD	23dBm±2.7dB	＜-40dBm	Compliant with Cat.1 protocol in 3GPP Rel-13 and Rel-14.
3	FDD	23dBm±2.7dB	＜-40dBm	
5	FDD	23dBm±2.7dB	＜-40dBm	
8	FDD	23dBm±2.7dB	＜-40dBm	
34	TDD	23dBm±2.7dB	＜-40dBm	
38	TDD	23dBm±2.7dB	＜-40dBm	
39	TDD	23dBm±2.7dB	＜-40dBm	
40	TDD	23dBm±2.7dB	＜-40dBm	
41	TDD	23dBm±2.7dB	＜-40dBm	

5.2.3 Sensitivity of reception

The receiving sensitivity of the NT26-FCN B series module is as follows:

Table 5-4 Conduction reception sensitivity under single transmission (throughput ≥ 95%)

Band	Network standard	Sensitivity (10M)	3GPP standard (10M)
1	FDD	-97.8dBm	-96.3dBm
3	FDD	-99.8Bm	-93.3dBm
5	FDD	-98.3dBm	-94.3dBm
8	FDD	-98.9dBm	-93.3dBm
34	TDD	-101dBm	-96.3dBm
38	TDD	-100.4dBm	-96.3dBm
39	TDD	-101.3dBm	-96.3dBm
40	TDD	-100.8dBm	-96.3dBm
41	TDD	-100.3dBm	-94.3dBm

5.3 Antenna reference circuit design

When using the NT26-FCN B series module, a π-type matching circuit needs to be added between the RF antenna interface of the module and the antenna interface of the user's baseboard. The typical antenna matching circuit and initial parameters are shown in the figure below:

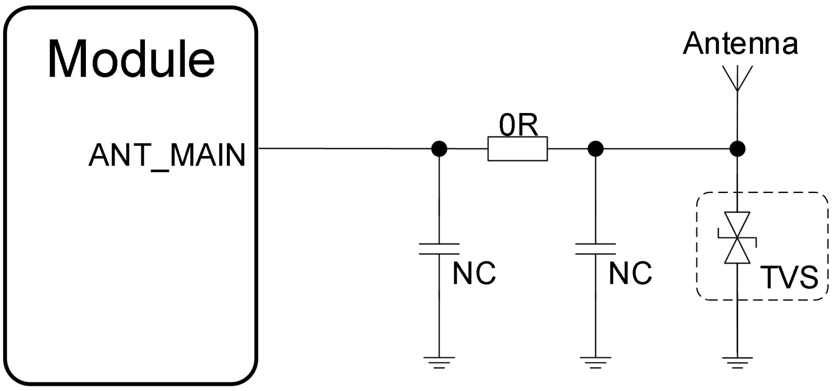


Figure 5-1 RF Antenna Reference Circuit

Notes

◆ The resistance uses 0 ohms, the capacitor position is not populated by default, and the recommended package for the components is 0201 or 0402.

◆ When using an external antenna, or when users can touch the antenna, it is recommended to reserve a TVS diode to enhance electrostatic protection. Because the parasitic capacitance of the TVS may affect the antenna performance, it is recommended to re-adjust the antenna after adding the TVS diode.

5.4 RF signal line wiring guide

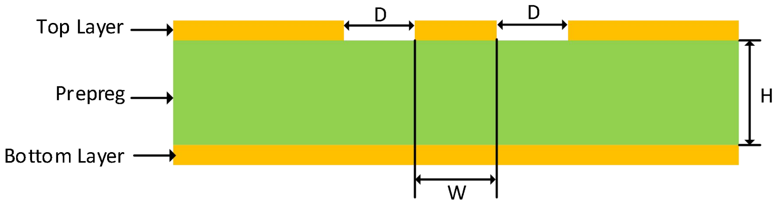
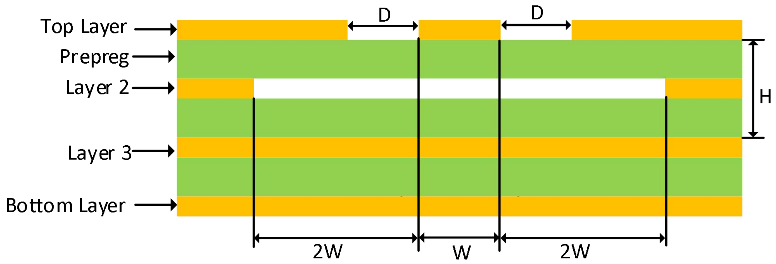
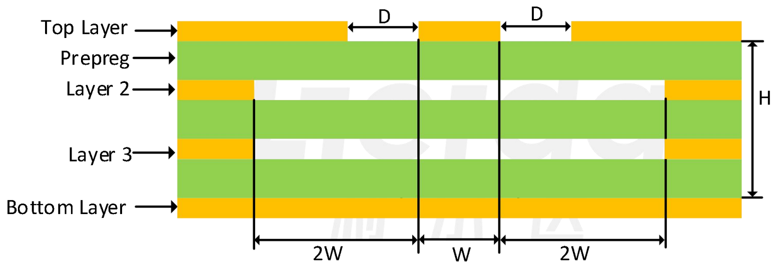
The wiring between the module antenna interface and the user antenna must meet the 50-ohm characteristic impedance requirements. At the same time, the RF wiring should be kept as short as possible to ensure minimal insertion loss of the RF wiring.

In general, the impedance of RF signal lines is determined by the material's dielectric constant Er, trace width W, spacing to GND D, and the thickness of the reference GND plane H.

In the field of IoT applications, the design of PCB characteristic impedance is usually achieved through coplanar waveguide to help the RF signal lines achieve better shielding and higher integration for small area design.

Common PCB coplanar waveguide designs include the following:

Table 5-5 Description of Common PCB Coplanar Waveguide Structures

Serial number	Pin name	Note
1		2-layer board Reference ground is at layer L2.
2		4-layer board Reference ground is at level L3. Layer 2 excavation
3		4-layer board Reference ground is L4 layer. Excavation of L2 and L3 floors.

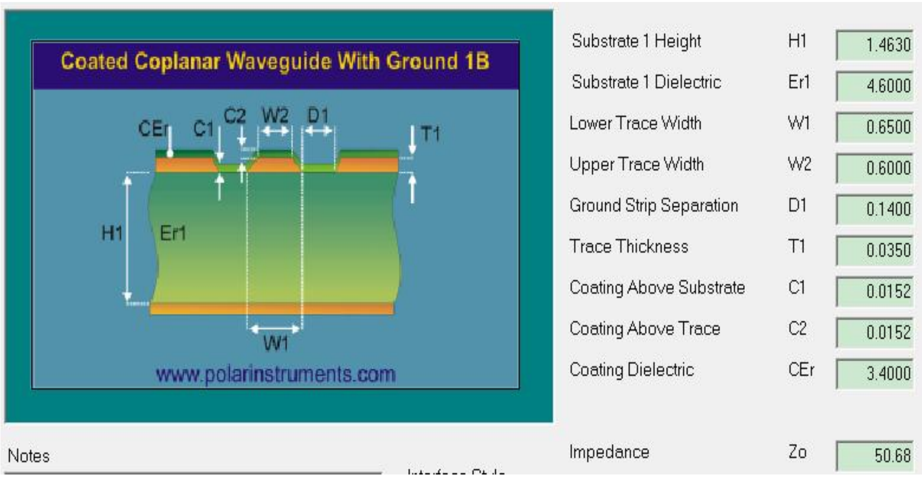
During the layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on actual conditions, appropriate line width, line spacing, and stack-up should be selected to achieve the best design effect.

Take the example of a 2-layer PCB with a thickness of 1.6mm to explain coplanar waveguide calculations.

◆ Choose the appropriate PCB substrate material, once the material is selected, the dielectric constant Er is basically determined (in this case, it is 4.6);

◆ Choose the appropriate line width, a good method is to select a line width that matches the pad size of the component to be soldered (note that it should not be too thin, in this case, choose 0.65mm).

◆ Based on the selected board thickness, line width, dielectric constant, etc., the spacing to ground can be roughly calculated. If the line spacing is not suitable, the line width can be adjusted to adjust the line spacing (calculated to be 0.14mm, basically appropriate).



Reference for the calculation method of 50 ohm impedance in Figure 5-2.

During the actual layout process, the following suggestions are provided for reference:

In the π-type circuit, three matching reserved components are placed closely near the antenna, while bypass components are recommended to be placed on both sides of the RF line.

◆ Irregular vias are placed on both sides of the RF trace GND plane, and there must be a complete GND plane below the entire RF trace space;

◆ There should be no other routing lines below the RF line to avoid affecting the RF performance or other circuits.

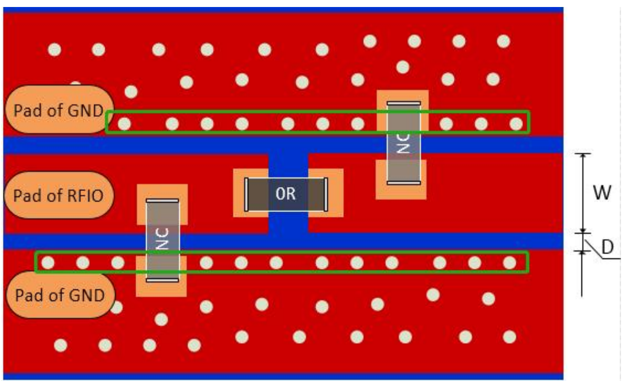


Figure 5-3 RF routing LAYOUT design schematic

The reasonableness of RF routing can be directly reflected in the module's conducted tests. In order to ensure the full performance of the product, cooperation with antenna design is also required. To better meet the requirements of antenna design, the following requirements are hoped to be achieved in PCB design. Below are guidelines for different layers of the whole PCB:

◆ When the product PCB is designed with 2 layers, it is best for both the TOP and BOTTOM LAYER directly below the module to be GND layers. The traces that need to be led out by the module should avoid passing directly below the module and should be led out from the outer side of the module.

◆ When the product PCB is designed as a 4-layer board, it is recommended that the traces to be routed from the module should be on the third or fourth layer, while reserving the first and second layers for the module as complete GND reference planes.

5.5 Antenna design requirements

The RF performance of the module is also affected by the antenna, and the antenna selection needs to meet the following requirements:

◆ Select antennas that are suitable for the operating frequency band of the module;

- Requires the characteristic impedance of the antenna to be 50 ohms to reduce losses at the RF cable and antenna connection;

◆ The lower the insertion loss within the operating frequency band, the better, such as VSWR ≤ 2, return loss ≥ 10dB, etc.;

Commonly used antennas suitable for Cat.1 scenarios are as shown in the following figure:

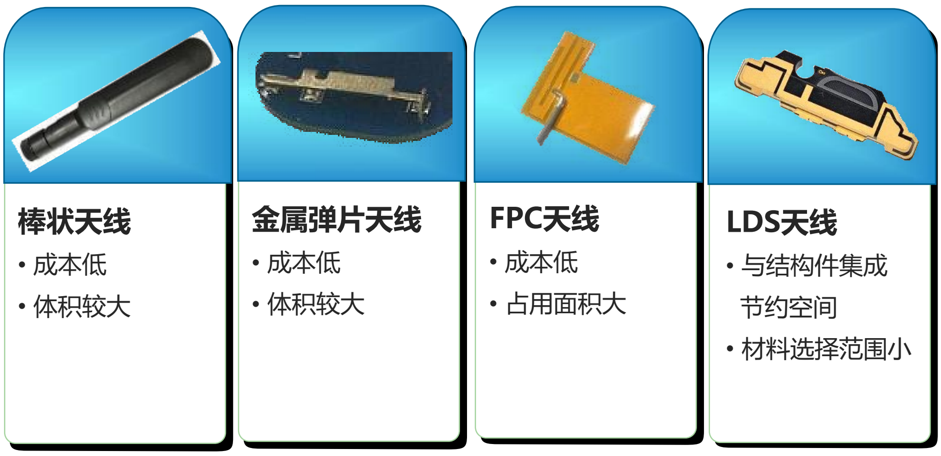
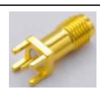


Figure 5-4 Commonly Used Antenna Types for Category 1

5.6 Recommended RF connectors

In some applications where customized antennas are needed, such as in environments where devices are shielded by metal casings, IPX or SMA connectors can be left on the motherboard to connect the antenna to the outside of the device via a coaxial cable to achieve better performance.

Table 5-6 Common RF Connectors

Serial number	Product Name	Image	Note
1	IPX socket		RF Line Connection
2	SMA connector		Can connect to RF cable or antenna.

6 Electrical performance and reliability

6.1 Absolute maximum rating

The description of the absolute maximum rated values of the NT26-FCN B series module interface is as follows:

Table 6-1 Absolute Maximum Ratings

Parameters	Description	Minimum value	Maximum value	Unit	Note
VBAT	Module supply voltage	-0.3	5.0	V	
VI_{AIO}	ADC input voltage	-0.3	3.6	V	
VI_{GPIO}	GPIO input voltage	-0.3	3.6	V	
Others	Other pin input voltage	-0.3	3.6	V	

Notes

When the operating conditions exceed the absolute maximum ratings, it may cause permanent damage to the module.

6.2 Rated power value

The electrical characteristics of the power interface of the NT26-FCN B series module are described as follows:

Table 6-2 Power Interface Electrical Characteristics

Parameters	Description	Minimum value	Typical value	Maximum value	Unit
VBAT	Module normal supply voltage(1)	3.3	3.8	4.5	V
VBAT	Module expansion power supply voltage (2)	3.1	3.8	4.5	V
USIM_VDD	USIM card power supply output	-	1.8/3.0	-	V

VDD_EXT	Reference power output	-	1.8	-	V
CAM_VDD	Camera power supply	-	1.8	-	V

Notes

◆ When the module operates within this voltage range, its performance meets the 3GPP standard requirements.

◆ When the module operates within this voltage range, it can still maintain normal operation without irreparable faults; only individual indicators, such as output power and other parameters, may exceed the range of the 3GPP standard. When the voltage returns to the normal operating range, all indicators of the module still comply with the 3GPP standard.

◆ Sleep mode: VDD_EXT will power off in sleep mode.

6.3 Power consumption

The power consumption of the NT26-FCN B series module is as follows:

Table 6-3 Module Power Consumption

Module description	Test conditions	Typical value	Unit
Shutdown mode	Module shutdown	0.97	uA
Sleep mode	AT+CFUN=0 (USB disconnected)	3/8.5	uA
	LTE-FDD @ PF = 32 (USB disconnected)	0.74	mA
	LTE-FDD @ PF = 64 (USB disconnected)	0.38	mA
	LTE-FDD @ PF = 128 (USB disconnected)	0.19	mA
	LTE-FDD @ PF = 256 (USB disconnected)	0.15	mA
	LTE-TDD @ PF = 32 (USB disconnected)	0.78	mA
	LTE-TDD @ PF = 64 (USB disconnected)	0.39	mA
	LTE-TDD @ PF = 128 (USB disconnected)	0.19	mA
	LTE-TDD @ PF = 256 (USB disconnected)	0.17	mA

Idle mode	LTE-FDD @ PF = 64 (USB disconnected)	3.15	mA
	LTE-FDD @ PF = 64 (USB connection)	23.53	mA
	LTE-TDD @ PF = 64 (USB disconnected)	3.09	mA
	LTE-TDD @ PF = 64 (USB connection)	23.54	mA
LTE data transmission	LTE-FDD B1	524	mA
	LTE-FDD B3	531	mA
	LTE-FDD B5	515	mA
	LTE-FDD B8	483	mA
	LTE-TDD B34	208	mA
	LTE-TDD B38	206	mA
	LTE-TDD B39	204	mA
	LTE-TDD B40	224	mA
	LTE-TDD B41	210	mA

Precautions

The sleep mode power consumption of the NT26FCNB0XXXX model is 8.5uA.

6.4 Digital logic voltage characteristics

The GPIO logic level definitions of the NT26-FCN B series module are as follows:

Table 6-4 Explanation of Digital IO Logic Levels

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Low level input	-	0.2*VDD_EXT	V
	VIH	High level input	0.7*VDD_EXT	-	V
Output	VOL	Low level output	-	0.15*VDD_EXT	V
	VOH	Output high level	0.8*VDD_EXT	-	V

Notes

The GPIO reference level of the module follows VDD_EXT, and VDD_EXT will lose power during sleep, so will the corresponding GPIO.

Power off.

The AGPIO logic level definition of the NT26-FCN B series module is as follows: (AON_VDD range: 1.7~1.9V; typical value: 1.8V)

Table 6-5 Explanation of Digital IO Logic Levels

Type	Parameters	Describe	Minimum value	Maximum value	Unit
Input	VIL	Low level input	-	0.2*LDO_AONIO	V
	VIH	High level	0.7*LDO_AONIO	-	V
Output	VOL	Low level output	-	0.15*LDO_AONIO	V
	VOH	Output high level	0.8*LDO_AONIO	-	V

Precautions

The AGPIO reference level of the module follows LDO_AONIO, and it will not lose power during sleep.

NT26-FCN B series module USIM card interface logic level definitions are as follows:

Table 6-6 USIM logic level description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Low level input	-	0.2*USIM_VDD	V
	VIH	Input high level	0.7*USIM_VDD	-	V
Output	VOL	Low level output	-	0.15*USIM_VDD	V
	VOH	Output high level.	0.8*USIM_VDD	-	V

Notes

The voltage level of USIM_VDD will be determined based on the detected category of the USIM card, supporting 1.8/3.0V USIM cards.

6.5 Static electricity protection

In life and production, static electricity is everywhere. Static electricity generated by human body static electricity, object friction, etc., may be transferred to the module through various ways, and may cause damage to the module. Therefore, it is necessary to pay attention to static electricity protection and take static electricity protection measures. For example:

◆ Wear anti-static gloves during the processes of research and development, production, assembly, and testing.

◆ When designing products, add electrostatic discharge protection devices at circuit interfaces and other points vulnerable to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-7 Electrostatic Protection Characteristics

Test interface	Discharge phenomenon	Note
VBAT	±8KV	Test standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	±8KV	
Antenna Interface	±8KV	
Others	-	

Static electricity protection detailed design reference "Lierda NT26-FCN B Series Hardware Reference Design Manual".

6.6 Work and storage temperature

Table 6-8 Operating Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Note
Normal operating temperature(1)	-35	+25	+75	℃	
Expand operating temperature(2)	-40	+25	+85	℃	
Storage environment temperature(3)	-40	+25	+90	℃	

Precautions

(1)When the module operates within this temperature range, the module's relevant performance meets the requirements of the 3GPP standard.

(2)When the module operates within this temperature range, the module can still maintain normal operation without irreparable faults; only individual indicators, such as output power and other parameter values, may exceed the range specified in the 3GPP standard. When the temperature returns to the normal operating range, all indicators of the module still comply with the 3GPP standard.

(3)This storage temperature range does not include packaging materials, and attention should be paid to the maximum temperature tolerance of the tape packaging.

6.7 Precautions

Modules need attention in the production and use process:

◆ When spraying the module, please try to avoid the spraying material flowing into the interior of the module;

◆ When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystals of the module.

◆ When producing and using modules, please avoid applying them in environments or enclosures containing any amount of mercury or mercury vapor, as this may lead to the risk of product failure or malfunction.

7 Mechanical dimensions

7.1 Mechanical dimensions

The mechanical dimension diagram of the module is as follows:

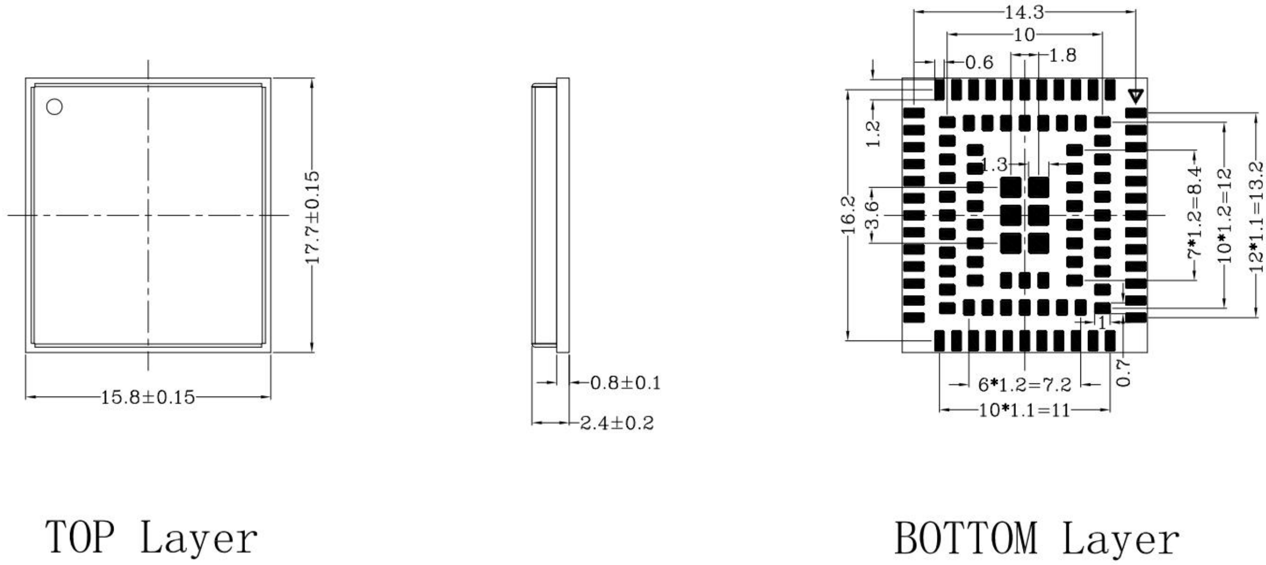


Figure 7-1 Module Mechanical Dimension Diagram

7.2 Module top view/bottom view

The module effect diagram is as follows, please refer to the actual module for details:

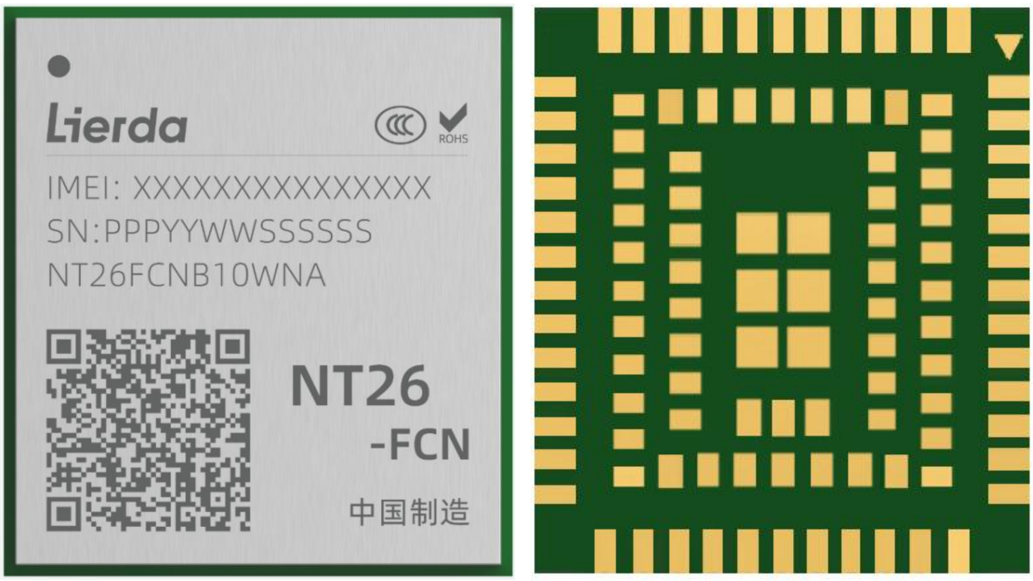


Figure 7-2 Module Top/Bottom View Schematic

7.3 Recommend encapsulation

The module recommends solder pads as shown in the following figure, and users can make minor adjustments according to their own production processes.

◆ The pins around the module are designed with rounded corners internally. When designing the bottom board for soldering, please consider using rounded corners for transition. The rectangular solder pads at the bottom of the module can be designed using the module pin dimensions. Please refer to the design diagram of a single solder pad in the figure below.

◆ To facilitate the opening of the ladder steel mesh, it is recommended not to layout other components within 2.0mm on the outer side of the module solder pad. Users can refer to the requirements of their own steel mesh manufacturers to determine this distance.

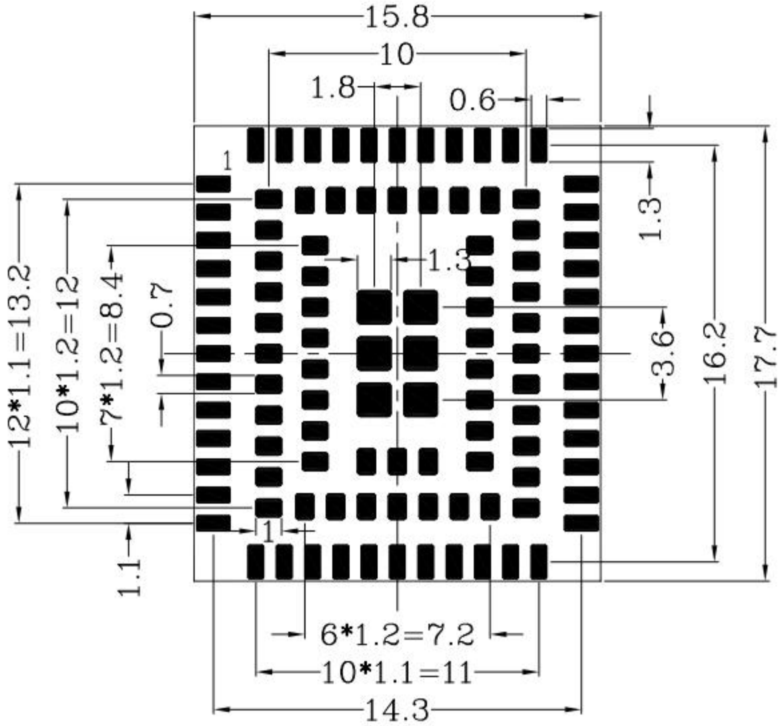


Figure 7-3 Module Recommended Solder Pads

8 Production and packaging information

This chapter describes information such as the patch process, packaging method, storage conditions of the module, which can help users better store and use this module.

8.1 Welding production

8.1.1 Furnace passing mode

During the SMT process, the following points need to be noted when reflow soldering:

If the bottom of the module is double-sided SMT, it is recommended to place the module in the second SMT process.

◆ It is best for customers' PCB to go through the reflow oven on the carrier tape during the first SMT placement, and also try to place them on the carrier tape for the second SMT placement.

◆ If, for special reasons, it is not possible to place it on the mesh belt for reflow soldering, consider using fixtures for reflow soldering on the track or placing a flat, high-temperature resistant straight template under the PCBA to prevent PCB deformation during reflow soldering, leading to virtual soldering of the module.

8.1.2 Reflow soldering operation guidance

PCBA reflow soldering temperature curve is related to the use of solder paste, and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operation, please refer to Figure 8.1 for lead-free reflow soldering operation guidelines.

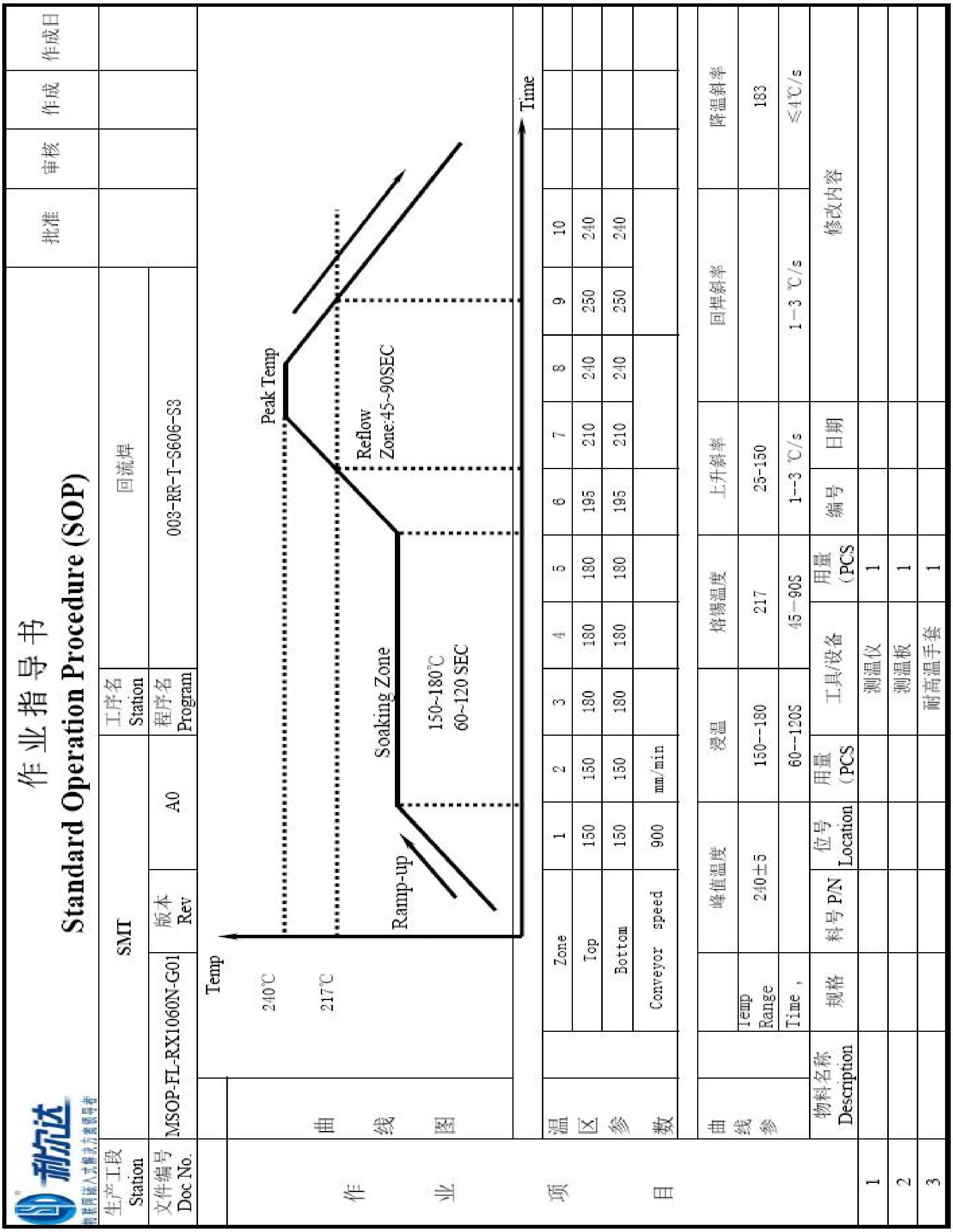


Figure 8-1 Lead-Free Reflow Soldering Operation Guide

8.1.3 Production process

During the production of welding or any process involving direct contact with the module, organic solvents such as alcohol, isopropanol, acetone, trichloroethylene, etc., must not be used to wipe the module shielding cover; otherwise, it may cause the shielding cover to rust.

If you need to spray or glue the module, please make sure that the spray or glue materials used will not chemically react with the module shield or PCB, and ensure that the spray or glue materials will not flow into the interior of the module.

8.1.4 Repair

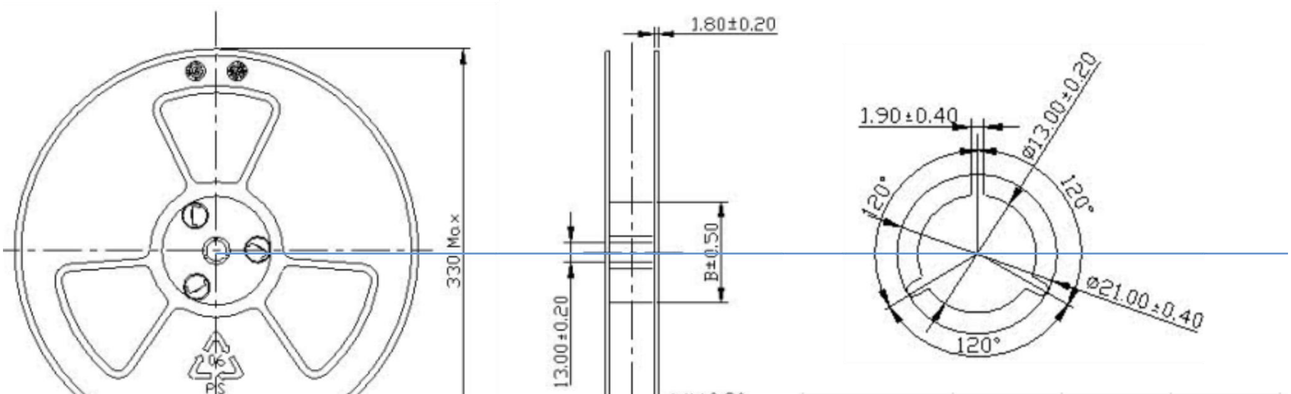
If the module has defects such as virtual welding or short circuit and needs to be repaired, please follow the parameters below:

◆ Lead-free process: Soldering iron temperature 380±10℃, soldering iron contact time ≤5S.

◆ Lead craft: Soldering iron temperature 350±10℃, soldering iron contact time ≤5S.

8.2 Packaging specifications

The factory packaging of this module is carried out using a rubber wheel method, with the reference dimensions of the rubber wheel as follows:



Reel Size(mm)			
Tape Width	B±0.5	W1±0.2	W2 Max
32	100	32.4	38.4

Figure 8-2 Belt Pulley Dimension Diagram

The loading direction is as follows: (Refer to the diagram, the actual content of the label shall prevail, pay attention to the position of module PIN1)

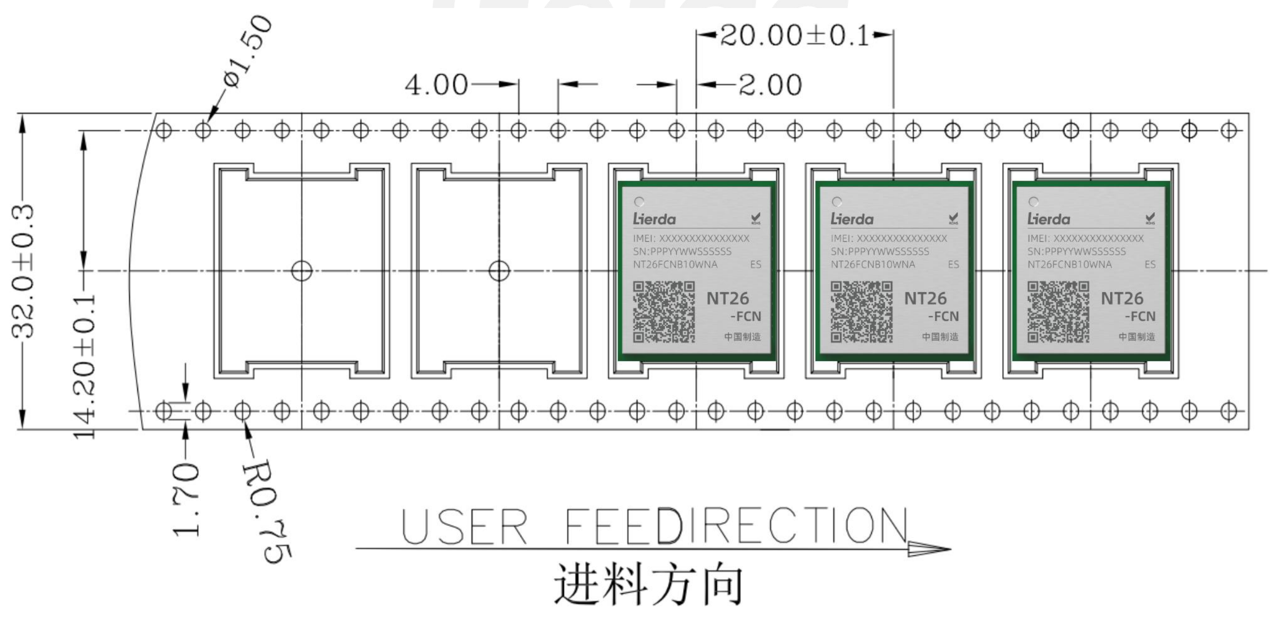


Figure 8-3 Loading Direction Indicator

8.3 Storage conditions

The module is shipped in the form of vacuum reel sealed bags, with Moisture Sensitivity Level (MSL) 3. Storage conditions are as follows:

◆ When the temperature is below 40°C and the humidity is below 90% (RH), the solderability can be guaranteed for 12 months in well-sealed packaging.

◆ After unpacking, ensure that surface mounting assembly is carried out within 168 hours under the condition that the ambient temperature is below 30°C and the relative humidity is below 60% (RH);

◆ If baking is required due to not meeting the above conditions, the tape and reel components should be baked at 60℃±5℃, humidity ≤60%RH for 48 hours; if accelerated baking is needed, the modules need to be individually baked after being taken out from the tape and reel (ESD protection should be observed during the extraction process), baked at 125℃±5℃, humidity ≤60%RH for 8 hours, with the total baking time not exceeding 96 hours.

Please refer to the IPC/JEDEC J-STD-033 standard for more detailed instructions.

9 Related documents and terminology abbreviations

9.1 Related documents

Reference documents are as follows:

Table 9-1 Related Documents

Serial number	Document Name	Annotation
[1]	Lierda NT26-FCN B Series Hardware Reference Design Manual	

9.2 Abbreviation of terms

The abbreviations and their corresponding explanations mentioned in this document are listed in the table below:

Table 9-2 Terminology Abbreviations

Abbreviation	Full English name	Chinese full name
3GPP	3rd Generation Partnership Project	The third-generation partner program
ADC	Analog-to-Digital Converter	Analog-to-Digital Conversion
ANT	Antenna	Antenna
ASM	Antenna Switch Module	Antenna switch module
DAC	Digital -to- Analog Converter	Analog-to-Digital Conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC Converter
DCXO	Digitally Controlled Crystal Oscillator	Numerically Controlled Crystal Oscillator

DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network coverage level
ESD	Electro-Static discharge	Static discharge
EOS	Electrical Overtress	Electrical super stress (surge)
ESR	Equivalent Series Resistance	Equivalent series resistance
EVK	Evaluation Kit	Assessment toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General input and output
I/O	Input/Output	Input and output interface
Imax	Maximum Load Current	Maximum current
Inorm	Normal Current	Normal (typical) current
bps	Bits Per Second	Units of speed
LCC	Leadless Chip Carriers	Leadless chip carrier package
LCM	LCD Module	LCD display module
LDO	Low Dropout Regulator	Low Dropout Linear Regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M protocol
MCU	Mirco Controller Unit	Microcontroller
MSL	Moisture Sentivity levels	Moisture Sensitivity Level
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed circuit board components
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy-saving mode

RF	Radio Frequency	Radio Frequency
RoHS	Restriction of Hazardous Substances	Limitation of harmful substances
RX	Receive	Receive
TAU	Tracking Area Update	Tracking area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient Voltage Suppression Diode
TX	Transmit	Send
UART	Universal Asynchronous Receiver & Transmitter	General asynchronous receiver and transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
URC	Unsolicited Result Code	Non-request result code
(U)SIM	(Universal) Subscriber Identification Module	Universal user identity recognition module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level

VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level